

DO LANGUAGE MODELS EXHIBIT INDUCTIVE REASONING IN JUDGEMENTAL FORECASTING TASKS?

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ABSTRACT

Language models produce accurate forecasts, but accuracy alone does not reveal whether they reason about how a situation works or rely on surface cues. Forecasting provides a demanding behavioral test: reliable predictions should use relevant evidence, infer relationships from limited observations, revise them when context changes, and transfer beyond familiar cases. Across real forecasting questions and controlled tasks¹, models distinguish direction-bearing evidence, use context to select predictive relationships, and benefit from training when mechanisms or events change. These gains extend to unfamiliar synthetic dynamics and held-out market events, although lower error does not always recover the hidden dynamics. Together, the results show that language models infer and reuse predictive structure when a task makes that structure recoverable.

1 INTRODUCTION

Recent language model (LM) systems approach human-crowd forecasting accuracy, and model ensembles can match it on some question sets (Halawi et al., 2024; Schoeneger et al., 2024). Forecasting agents have now been tested on Metaculus, Polymarket, and Kalshi, with mixed results (Lu, 2025; Cao et al., 2026; Zhang et al., 2026). Their growing ability motivates our question: *why are language models good at forecasting?*

A forecast connects limited evidence to an outcome through assumptions about how a process behaves. Figure 1 illustrates the distinction: after two heads, the next-flip probability remains 0.5 if the coin is known to be fair, but rises when the carnival setting makes a rigged coin plausible. Real forecasts require similar judgments: whether news matters depends on what it changes, how long its effect lasts, and how that effect reaches the outcome (Mellers et al., 2014; Tetlock & Gardner, 2016).

The same probability can result from retrieving a similar case, mapping an analogy, copying a market, or applying a learned model of the process. Prompted rationales raise accuracy on some reasoning tasks (Kojima et al., 2022), but a verbal rationale cannot distinguish these possibilities, because generated explanations need not identify the cause of a model’s output (Jacovi & Goldberg, 2020; Turpin et al., 2023). We

FORECASTING WITH A WORLD MODEL



Figure 1: After observing two heads in a row at a carnival coin-flipping game, forecasts differ depending on whether the coin is treated as fair or potentially rigged. We asked GPT-5.4, Claude Opus 4.8, and DeepSeek V4-Pro to forecast the next-flip question shown. Each logo marks that model’s mean over eight temperature-0.7 forecasts: .644, .681, and .712, respectively (Appendix B.1).

¹ 🤖 Study dataset and artifacts. Link anonymized for review.

therefore ask: *Do language models' forecasts exhibit behavioral signatures of inductive reasoning over learned world structure?* Here, **judgemental forecasting** converts incomplete, heterogeneous evidence into probabilities about future events. We define *inductive reasoning* as inferring a predictive relationship from context and observations, revising that relationship with evidence, and applying it to a new case (Tenenbaum et al., 2006; 2011; Griffiths et al., 2010; Lake et al., 2015; 2017). *World structure* refers here to predictive relationships among variables, not to a specified internal representation.

1.1 EMPIRICAL APPROACH

We test three requirements of inductive forecasting:

- **Selective updating (Experiment 1).** On unresolved real questions, we intervene on evidence direction and relevance. Across nine models and 30,094 updates, forecasts follow the packet's stated direction and move $1.9\text{--}24\times$ farther under directional packets than under topically matched orthogonal ones. The test rejects indiscriminate topic reactivity, but a model can pass by classifying the packet's stated implication.
- **Context-guided response inference (Experiment 2).** In Coin City, matched prompts isolate a qualitative cue that links a target city to a response regime. Adding that one sentence raises the correlation between forecast-implied and latent responsiveness from $-.07\text{--}.08$ to $\rho = .52\text{--}.70$ with every number held fixed, and inverting it inverts the assignment. Cue inversion and accumulating observations reveal whether the model uses and overrides that link; a deterministic analogical baseline tests whether explicit mechanism recovery is needed.
- **Transfer after training (Experiments 3 and 4).** We train forecasting policies and extend Coin City by crossing an unseen persistent mechanism with a new surface domain. Rewarding episode-specific relationships beats a matched population-prior control by .32 [.13, .49] MAE under joint domain-and-mechanism shift, and training on historical markets lowers sealed-test Brier score from .1316 to .1159 over 1,024 held-out events. A second synthetic study evaluates broader unseen mechanism compositions. Matched rewards distinguish episode-specific relationships from a population response, while structureless and untrained policies diagnose format learning and base-rate improvement.

Together, these interventions test whether forecasts track changes in evidence, context, and the governing relationship. That pattern goes beyond outcome accuracy and is expected if models infer and reuse predictive structure.

2 RELATED WORK

Judgemental and language-model forecasting. Research on judgemental forecasting separates accuracy from process. Strong human forecasters use base rates, decompose questions, seek diverse evidence, and update frequently; unaided expert judgment and manual adjustments can remain miscalibrated or biased (Tetlock, 2005; Mellers et al., 2014; Tetlock & Gardner, 2016; Lawrence et al., 2006). Autocast and ForecastBench provide time-indexed, leakage-controlled evaluations of language models (Zou et al., 2022; Karger et al., 2025). Retrieval-augmented generation, aggregation, and ensembling have brought some systems close to crowd accuracy (Lewis et al., 2020; Halawi et al., 2024; Schoenegger et al., 2024), while tournament entries report more mixed real-world performance (Schoenegger & Park, 2023). These studies measure forecast quality; we test how forecasts respond to evidence, examples, and training.

In-context inference and analogy. Transformers can infer unseen functions from input–output examples, with success depending on the training distribution (Garg et al., 2022; Chan et al., 2022). Language models also solve some abstract analogy tasks, and analogical prompting can improve reasoning (Webb et al., 2023; Yasunaga et al., 2024). Because either relationship inference or analogical matching could explain Coin City performance, we compare model forecasts with a deterministic, context-conditioned analogical regression.

World structure and behavioral evidence. World models represent environmental dynamics for prediction or control (Ha & Schmidhuber, 2018; LeCun, 2022), and sequence models can encode task state and support predictions about agents (Li et al., 2023; Andreas, 2022). Predictive success can rest on shortcut features and need not generalize to a new task or structural change (Geirhos

et al., 2020; Mitchell, 2023; Vafa et al., 2025). Causal formalisms make such structure explicit (Pearl, 2009; Peters et al., 2017), and formal benchmarks test queries over specified graphs (Jin et al., 2023); we test whether forecasts behave as if a model inferred and reused predictive relationships when no graph label is requested. Because natural-language rationales may not faithfully describe the computation behind an answer (Jacovi & Goldberg, 2020; Turpin et al., 2023), we use behavioral interventions that specify expected invariances or directional changes (Ribeiro et al., 2020; Elazar et al., 2021).

3 METHODS

We use controlled simulations and prospective and historical Polymarket questions (Gentzel et al., 2019; Polymarket, 2026). The simulations provide known data-generating processes, allowing us to change one source of information at a time. The market studies test evidence use on open-ended questions and accuracy on held-out resolved events. Contrasts are paired by question or episode. For training experiments, uncertainty is first estimated across training seeds and then across episodes within seed.

3.1 SYNTHETIC STUDIES

3.1.1 COIN CITY: INFERRING A RESPONSE REGIME

Each Coin City episode describes three cities with stable but unobserved poll responses to news. City A and City B provide reference cases. City C provides up to four observations and a qualitative description of its news environment. The description indicates which response regime is more likely without stating a coefficient.

We generate 250 independent episodes. For city j in episode i ,

$$\Delta p_{ij} = g_j x_{ij} + \epsilon_{ij}, \quad \epsilon_{ij} \sim \mathcal{N}(0, 5^2), \quad (1)$$

where x_{ij} is signed net news. Strong cities have $g_j \sim \mathcal{N}(0.90, 0.03^2)$ and weak cities have $g_j \sim \mathcal{N}(0.25, 0.03^2)$. City A and City B each provide four reference cases arranged as matched equal-and-opposite news pairs. A query asks for City C’s ending poll after news $x \in \{-10, \dots, -4, 4, \dots, 10\}$, given $k \in \{0, 1, 2, 3, 4\}$ City C cases.

Four prompt arms vary the evidence available to the model: target cases alone; reference cases without City C context; references with the correct City C context; and references with the opposite context label. At $k = 0$, the correct-context contrast tests whether the label selects a reference regime, and the opposite-label arm tests whether reversing the label reverses that selection. Increasing k tests whether City C observations reduce the label’s influence. Figure 2 shows the shared task, reference cases, and information arms.

Claude Opus 4.8, Claude Opus 5, GPT-5.6, DeepSeek-V4-Pro, Kimi K3, and Gemini 3.6 Flash are evaluated on every frozen episode, evidence depth, and arm: 30,000 attempts in total. Forecast MAE is measured against the noise-free expectation $p_0 + g_C x$. The forecast-implied response is $\hat{g} = (\hat{p} - p_0)/x$. We report its MAE and across-episode correlation with g_C . A deterministic analogical baseline selects the label-matched reference city, pools its cases with visible City C cases, and fits a through-origin OLS slope; the opposite-label arm selects the other city. Paired intervals use 2,000 episode bootstrap samples (Efron, 1979).

Shared prompt in every arm

Each row is a separate polling case, not a time series. Positive net news favors the candidate and negative net news harms the candidate. Poll change is the ending poll minus the starting poll. Within a city, typical responsiveness to net news is stable across cases, although individual end polls remain noisy. Rows with the same Pair label have equal-sized news in opposite directions.

Query: A new City C case begins with a poll of 53.9 and has net news +5. What poll do you predict at the end of this new case? Think carefully about the expected poll, but do not provide step-by-step working. Respond with one JSON object only: {"rationale": "one short sentence", "predicted_poll": <number from 0 to 100>}

Illustrated arm: References + target context**CITY A**

Background: Residents generally encounter campaign developments through **national news coverage**, where campaign stories receive sustained attention and tend to produce larger immediate poll movements.

Pair	Start	News	Delta	End
1	45.4	-8	-6.5	38.9
1	45.4	+8	+12.2	57.6
2	55.1	-6	+0.9	56.0
2	55.1	+6	+11.4	66.5

CITY B

Background: Residents generally encounter campaign developments through **local news coverage**, where campaign stories compete with local issues and tend to produce smaller immediate poll movements.

Pair	Start	News	Delta	End
1	35.0	-10	-5.6	29.4
1	35.0	+10	+9.3	44.3
2	51.8	-10	-5.5	46.3
2	51.8	+10	+7.2	59.0

CITY C - TARGET CONTEXT

Background: City C residents generally encounter campaign developments through **national news coverage**.

Pair	Start	News	Delta	End
1	57.8	+8	+9.4	67.2
1	57.8	-8	-1.8	56.0
2	47.4	+10	+8.1	55.5
2	47.4	-10	-10.6	36.8

Figure 2: **Coin City prompt and information arms.** Every arm receives the instructions and query shown. The illustrated episode ($k = 4$) additionally provides City A/B references and City C’s qualitative context. The no-context and opposite-context arms change only the City C sentence.

3.1.2 COIN CITY: TRAINING ON A DIRECT RESPONSE, TESTING A HIDDEN MEDIATOR

Training uses the direct, one-period relationship in Equation 1. Every prompt shows two complete 14-case reference systems with their background descriptions, a target background with k observed cases, and a query for ten shock-horizon outcomes. The causal and population-prior arms use byte-identical prompts and the same multiset of answer keys. Causal training pairs episode i with its own key y_i^* ; the population-prior control pairs it with $y_{\pi_k(i)}^*$ under a fixed derangement within evidence depth k . The latter preserves the population response distribution but breaks the link to the episode’s references and target history. A structureless diagnostic scrambles the displayed drivers and rewards a flat response; the untrained base receives the same evaluation prompts without training. Table 24 in Appendix C.1 shows all four conditions on one episode.

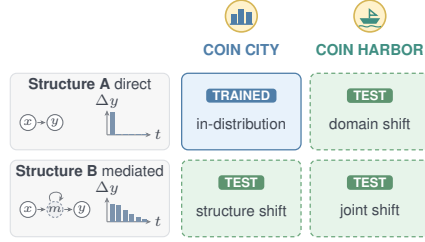


Figure 3: **Transfer grid.** Training uses the shaded cell only; all four are evaluated. Bars are each mechanism’s response Δy over the periods t following one shock: A returns to baseline at once, B builds through the unobserved mediator m and persists.

Policies train only on Coin City under Structure A, the direct mechanism $x_t \rightarrow y_{t+1}$. Figure 3 crosses this trained cell with a new surface domain (Coin Harbor), the mediated Structure B ($x_t \rightarrow m_{t+1} \rightarrow y_{t+1}$), and both shifts together.

The evaluation crosses each cell with a correct, absent, or misleading pathway hint and $k \in \{0, 2, 4, 8\}$ target observations, yielding $2 \times 2 \times 3 \times 4 \times 30 = 1,440$ paired prompts; each training run contains 4,800 Coin City/Structure A prompts. Equations, parameters, mediator names, and structure labels are never shown, while the two complete references make both response patterns observable. The primary estimand is population-prior minus causal response MAE in the joint-shift cell under the correct hint. The analysis uses the three complete Qwen3-4B seeds per arm at round 300 under greedy and five-draw stochastic decoding. Draws are averaged within task before a paired seed-first bootstrap; the stochastic endpoint is primary. The registered roster also includes Qwen3-8B and Llama-3.1-8B. Verbatim pathway descriptions and full parameters appear in Appendix C.1; the separate broad-composition study is in Appendix C.3.2.

3.2 POLYMARKET STUDIES

3.2.1 PROSPECTIVE EVIDENCE INTERVENTIONS

We sample 100 unresolved binary markets covering politics, government, elections, economics, and geopolitics. Each market is open, 14–180 days from resolution, and has at least 1,000 central limit order book (CLOB) transactions; none had resolved at collection. We report nine models: three frontier API systems (OpenAI, 2026; Anthropic, 2026; DeepSeek-AI, 2026) and six scale-valid Qwen 2.5 or Llama-family open-weight models (Qwen Team, 2025a; Grattafiori et al., 2024). The appendix retains an additional mixed-scale local deployment for directional-compliance auditing (Appendix A.3.1).

Agents first return a structured event model, probabilities for H_1 (YES) and H_2 (NO), and the headline forecast $\hat{p} = P(H_1)$. We then disable web search and present one generated evidence packet at a time. For each market, three packets support H_1 , three oppose it, and three are topically related but designated orthogonal. The main analysis contains 30,094 valid updates. Most directional packets state their implication, so success does not require a multi-step causal derivation.

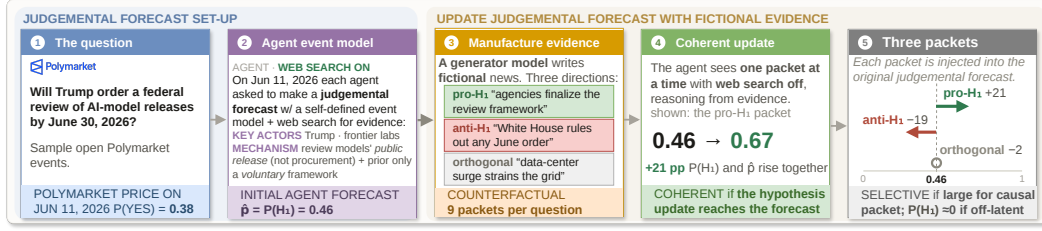


Figure 4: **Prospective evidence-intervention protocol.** An agent forecasts an unresolved question, then receives one generated evidence packet with search off. Directional correctness asks whether the update follows a pro- or anti- H_1 packet. Selectivity compares movement under directional and orthogonal packets. The output contract requires $\hat{p} = P(H_1)$, so agreement between those fields is partly schema compliance.

Sensitivity, the primary comparison, divides mean absolute movement under directional packets by movement under orthogonal packets. The results table also reports these constituent means as D/O (directional/orthogonal), in percentage points. **Evidence-Hypothesis Consistency (EHC)** is a secondary directional check: the correctness rate for stated $P(H_1)$ among updates of at least 3 percentage points. We also report unconditional correctness and thresholds of 0, 1, 3, and 5 points. Table colors use four descriptive bands. Sensitivity and D/O cells are red below $2.5\times$, orange from $2.5\times$ to below $5\times$, yellow from $5\times$ to below $10\times$, and green at $10\times$ or above; D/O is colored by its ratio. EHC cells are red below .90, orange from .90 to below .95, yellow from .95 to below .98, and green at .98 or above. The appendix reports a separate output-compliance check for the headline forecast field.

A separate human study checks the synthetic materials. Seven completed, quality-eligible reviewers judged the same 18 balanced packets (six pro-YES, six anti-YES, and six orthogonal), seeing the market question, resolution criteria, and frozen June 10 background. They rated update direction, confidence, clarity, plausibility, and premise usability. Reviewers matched the registered direction on 96/126 judgments (76.2%), and their majority label matched it on 16/18 items ($\kappa = .84$). Exact agreement was 29/42 for pro- H_1 , 33/42 for anti- H_1 , and 34/42 for orthogonal packets. These figures include every eligible completed submission; one reviewer in the nine-person extension has not submitted ratings. Appendix A.5 reports per-reviewer and per-item judgments; Appendix A covers prompts, filters, deployments, and missingness.

3.2.2 HISTORICAL-MARKET TRAINING AND SEALED EVALUATION

Resolved political, governmental, macroeconomic, and geopolitical markets are cut off 30 days before their earliest archived terminal timestamp. Prompts retain only the question, rules, cutoff, and genuine pre-cutoff prices. Outcomes and post-cutoff fields are withheld. Calendar splits contain 1,736 training, 512 development, and 1,024 test markets, with no event or normalized question family shared across splits.

Qwen3-4B (Qwen Team, 2025b) is trained with rank-32 LoRA (Hu et al., 2022) and Dr. GRPO (Liu et al., 2025) for 300 updates (4,800 prompt presentations) at each of three seeds. Reward is the proper score $1 - (\hat{p} - y)^2$ (Brier, 1950); training-time evaluation uses development data only. The sealed test compares the base and three adapters with the contemporaneous market, training prevalence, and train-only logistic recalibration (Dawid, 1982). Brier and log-loss intervals resample connected event/template families (Brier, 1950; Good, 1952). A secondary update analysis fixes one initial forecast and all 900 evidence packets across base and trained models, disables search, and decodes greedily. Appendix C.2 gives the censoring, checksum, and optimization details.

4 RESULTS

4.1 EXPERIMENT 1: SELECTIVE UPDATING ON UNRESOLVED QUESTIONS

Experiment 1 tests whether forecasts respond to the stated direction and relevance of new evidence. Because most directional packets state their implication, a model can pass without deriving a causal path. It must still distinguish directional evidence from topically related controls.

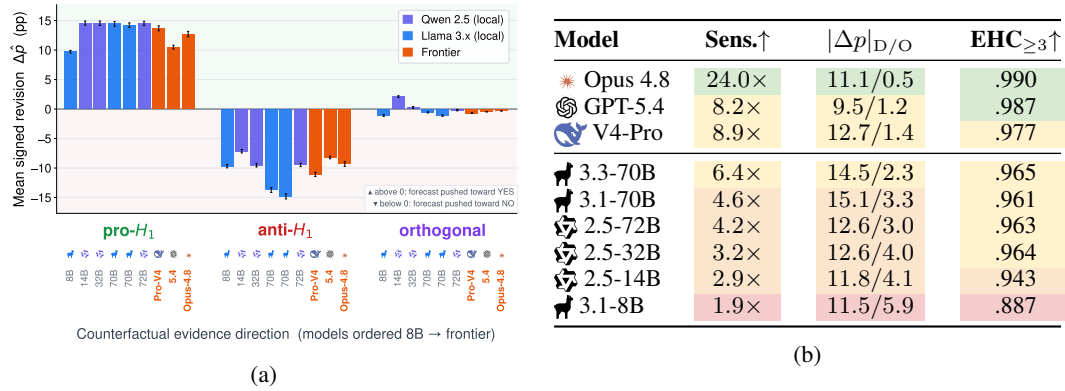


Figure 5: **Direction-selective updating across nine models and 30,094 valid update records.** (a) Mean signed forecast revision by counterfactual direction; error bars are standard errors. (b) Packet selectivity and directional correctness. Sensitivity is mean directional divided by mean orthogonal absolute revision; D/O reports those means in percentage points. EHC is the conditional directional-correctness rate among movements of at least three percentage points. Colors progress from red through orange and yellow to green using the descriptive bands defined in Methods. Output-field compliance and a 0/1/3/5-pp threshold sweep appear in Appendix A.3.4.

Models distinguish direction-bearing packets from topical controls. Frontier sensitivity ranges from 8.2× to 24.0×: the same agents move farther under pro- and anti- H_1 packets than under orthogonal packets. Every scale-valid local model shows the same ordering, with sensitivity from 1.9× to 6.4×. The result is therefore not confined to the frontier systems, although selectivity is weaker in the smaller open-weight models.

Directional movements usually follow the stated sign. Across 30,094 valid updates, conditional EHC ranges from .887 to .990 (Figure 5), and all three frontier agents exceed .97. Their unconditional directional correctness is 97.3–98.4%; the 3-point conditional metric omits 6.9–18.1% of directional updates that move less than the threshold. Because this measure is near ceiling, it serves as a directional guardrail rather than the primary model comparison.

These results reject an update policy that reacts equally to any news-like text about the market. They remain compatible with reading the packet’s explicit polarity and applying a learned numerical shift. Experiment 2 instead makes the qualitative cue informative only through numerical reference cases.

4.2 EXPERIMENT 2: CONTEXT-GUIDED RESPONSE INFERENCE IN COIN CITY

Experiment 2 tests whether models use a qualitative cue to determine which numerical relationship applies to a new target. Reference cases define the cue’s meaning. Matched correct- and opposite-context prompts test whether that meaning changes the forecast.

Context links City C to a response regime. With no City C observations ($k = 0$), adding its context lowers MAE relative to the matched no-context arm for every deployment. Improvements range from 0.37 to 0.92 poll points, with all bootstrap intervals excluding zero; the cross-model median falls from 2.59 to 1.85. All numeric inputs are identical in the two arms. Forecast-implied responses correlate .52–.70 with City C’s true responsiveness, compared with −.07 to .08 without context. By $k = 4$, the MAE advantage narrows to 0.12–0.23 because City C’s own cases reveal the relationship.

Reversing the cue reverses regime assignment. At $k = 0$, the opposite label raises MAE to 4.20–4.36, 1.52–1.76 points above no context, and response correlations become −.50 to −.68. As City C observations accumulate, the wrong-label penalty falls to −0.01–+0.38 points at $k = 4$. Models therefore use the cue when evidence is sparse and reduce its influence when observations disagree.

The context-conditioned analogical baseline reproduces the pattern. The deterministic label-matched regression baseline obtains MAE 1.84 and $\rho = .66$ at $k = 0$, inside the deployment ranges of 1.60–2.47 and .52–.70. With the label inverted, it obtains MAE 4.56 and $\rho = −.67$. Coin City

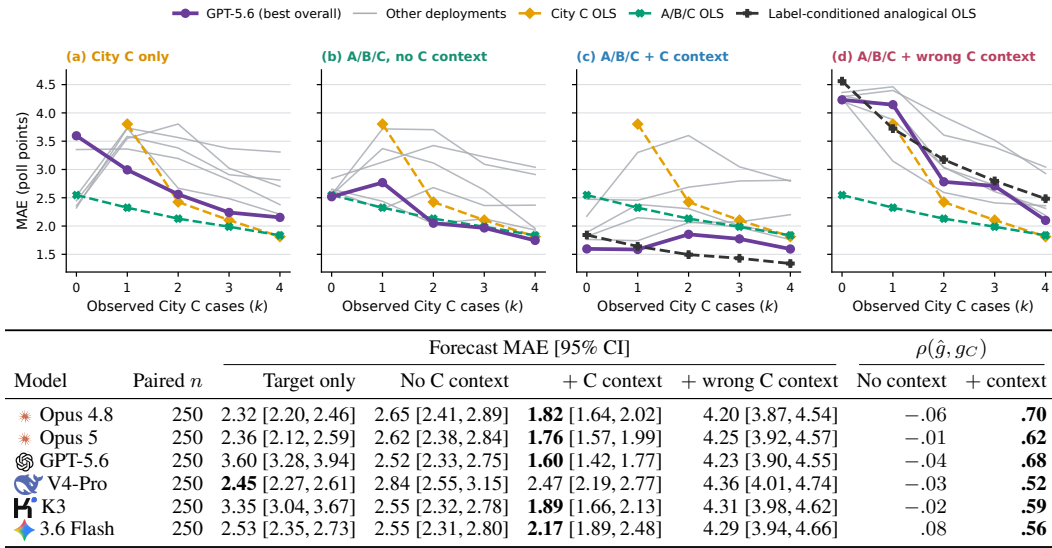


Figure 6: **Context improves zero-shot forecasts and response-regime assignment across six deployments.** The plots show forecast MAE as City C observations accumulate. Panels successively add references, the correct target context, and a misleading context naming the opposite regime. Gray lines are deployments, purple is the best overall deployment, and dashed curves are deterministic OLS benchmarks. The table gives per-model MAE and latent-slope correlation at $k = 0$, with 95% bootstrap intervals over 2,000 episode resamples. Paired contrasts are in Appendix B.6.

therefore shows contextual assignment, cue inversion, and evidence override, but the analogical baseline reproduces the main pattern. Experiment 3 asks whether training on this direct response improves forecasts in a new domain governed by an unseen persistent mediator.

4.3 EXPERIMENT 3: DOMAIN AND STRUCTURAL TRANSFER IN COIN CITY

Experiment 3 asks whether training on Coin City’s direct response transfers to a new surface domain, an unseen mediated mechanism, or both (Figure 3). The model never sees equations, parameters, mediator names, or structure labels; complete reference systems and qualitative descriptions provide the available pathway evidence (Appendix C.1).

We report the registered round-300 Qwen3-4B endpoint under greedy and five-draw stochastic decoding. All nine trained runs and the untrained base use the same 1,440-task evaluation. At the analysis freeze, the Qwen3-8B and Llama-3.1-8B confirmatory grids were incomplete, so no cross-model estimate is reported.

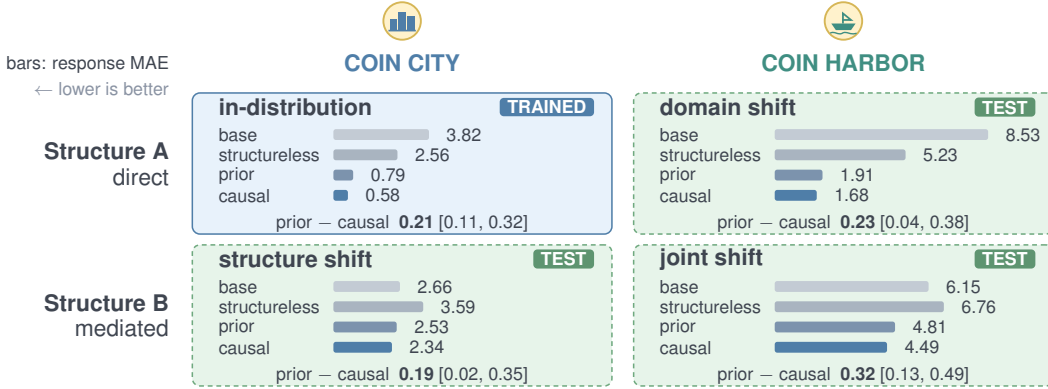


Figure 7: **Qwen3-4B Coin City transfer at round 300 under five-draw stochastic decoding**, on the design grid of Figure 3. Each bar is response MAE after averaging five draws within task and four evidence depths; shorter is better, and trained arms average three seeds. The paired seed-first prior-minus-causal difference is printed under each cell, in bold where its 95% interval excludes zero; positive values favor episode-specific reward.

Episode-specific reward improves the hardest transfer forecast. Under five-draw stochastic decoding, joint new-domain/new-structure MAE is 6.15 for the untrained base, 6.76 for structureless training, 4.81 for population-prior training, and 4.49 for causal training. The paired prior-minus-causal difference is .319 [.134, .489] (Figure 7). Causal training has the lowest point estimate in all four cells, and all four intervals exclude zero. Draws are averaged within task before the seed-first bootstrap.

Greedy decoding agrees directionally. Its joint-cell MAEs are 6.09, 6.69, 4.77, and 4.59, respectively, with a prior-minus-causal difference of .174 [.039, .316]. Causal training again has the lowest point estimate in all four cells; the interval excludes zero for the in-distribution and joint-transfer cells, but not for the domain-only or structure-only cells. The complete greedy grid appears in Appendix C.1.5.

Lower error does not coincide with recovery of the unseen dynamics. In the joint cell, correct, absent, and misleading hints yield stochastic causal MAE of 4.49, 4.64, and 4.81, so qualitative-hint separation remains small. The greedy values are 4.59, 4.68, and 4.86. A nearest-pathway response-shape diagnostic classifies 0.0% of stochastic causal draws and 2.5% of greedy causal forecasts as Structure B, compared with 87.3% and 81.7% for the corresponding, less accurate base forecasts. Thus, episode-aligned reward improves intervention forecasts, including under joint transfer, but the result remains compatible with conservative response shrinkage rather than reconstruction of the persistent mediator.

A complementary 48-endpoint study evaluates four held-out blocks spanning unseen topologies, nonlinearities, mixed mechanism compositions, and parameter extrapolation (Appendix C.3.2). Its full Qwen3-4B, Qwen3-8B, and Llama-3.1-8B factorial is heterogeneous: causal training beats the matched population-prior arm in both Qwen3-4B repeated-draw cells and the hidden Llama cell, loses in the disclosed Qwen3-8B cell, and is unresolved elsewhere. Yet it beats the untrained base under repeated sampling in all six model-by-disclosure cells. Showing the mechanism does not reliably change the causal-over-prior contrast. Coin City remains the main analysis because it directly extends Experiment 2.

4.4 EXPERIMENT 4: TRANSFER TO HELD-OUT HISTORICAL MARKETS

Experiment 4 tests transfer on resolved events. Training, development, and sealed test sets share no event families or normalized question templates. The test contains 1,024 markets, and all 4,096 base and trained forecasts parse.

Table 1: **Historical-market training improves sealed-test accuracy.** The test contains 1,024 markets. Lower is better; negative differences favor the trained model.

Forecaster	Updates	Brier ↓	Δ base	Δ market	Log loss ↓
Untrained Qwen3-4B	0	.1316	—	+.0179	.5114
Trained, seed 42	300	.1142	-.0174	+.0005	.3780
Trained, seed 43	300	.1140	-.0176	+.0003	.3772
Trained, seed 44	300	.1194	-.0122	+.0058	.4330
Trained mean	300	.1159	-.0157	+.0022	.3960
Contemporaneous market	—	.1137	-.0179	—	.3658
Market + train-fit logit	—	.1123	-.0193	-.0014	.3605

Every training seed improves on the untrained model. Mean Brier score falls from .1316 to .1159, a difference of $-.0157$ [$-.0253, -.0067$]; log loss falls from .5114 to .3960. Exact event or question-template repetition across splits cannot explain the gain.

The trained model is statistically indistinguishable from the market. Training closes 88% of the base-to-market Brier gap. Its difference from the contemporaneous market is .0022 [$-.0026, .0072$] and from train-only logistic recalibration is .0036 [$-.0013, .0087$]. Both intervals include zero.

Training improves directional consistency; its effect on selectivity is unresolved. On fixed initial forecasts, training raises directional consistency by .0101 [$.0004, .0207$]. The sensitivity difference, with interval $[-.07, .48]$, is unresolved. The historical study shows accuracy transfer relative to the untrained checkpoint. It does not show performance beyond market prices or a general change in evidence selectivity.

5 LIMITATIONS

Behavior does not identify internal computation. Experiment 1 is compatible with semantic polarity classification followed by a learned update size. A label-conditioned analogical regression matches Coin City. In Experiment 3, lower forecast error coexists with near-chance mechanism labels. The experiments establish specific input–output behavior but remain compatible with several internal algorithms and representations.

The evaluated environments remain limited. The prospective evidence packets are generated and usually state their direction; the seven-reviewer packet ratings fall short of the registered agreement thresholds, and the nine-reviewer extension remains incomplete. Experiment 2 uses two well-separated linear response regimes and a cue that is correct or inverted; its arbitrary-symbol control covers one deployment. Experiment 3 trains on one direct Coin City family and evaluates one held-out mediated family in two surface domains. Its hints are correct, absent, or misleading, but remain categorical descriptions from a fixed template. The appendix composition study adds unseen non-linear and temporal compositions and parameter extrapolation, but every world comes from a finite catalog and shared prompt format. We do not test graded cue reliability, ambiguous mechanisms, or new sets of variables.

The Coin City model comparison remains incomplete. The Coin City result covers Qwen3-4B at the registered round-300 checkpoint under both greedy and five-draw stochastic decoding, but its Qwen3-8B and Llama-3.1-8B endpoints were incomplete when this analysis was finalized. The separate broad-composition study includes all three models and finds heterogeneous causal-versus-prior effects rather than a roster-wide advantage. Both studies use one LoRA/Dr. GRPO recipe and three seeds per arm, and five draws do not characterize the full output distribution.

The market studies have separate validity constraints. The unresolved-market interventions cannot be scored against outcomes, and their evidence packets are hypothetical. Historical markets use one 30-day cutoff and selected binary domains. Pretraining exposure cannot be audited, archived text does not include edit histories, and the update analysis fixes the initial forecast. Historical train-

ing improves on the base model but is not statistically distinguishable from contemporaneous market prices.

6 CONCLUSION

Across four studies, language models use evidence selectively, infer context-dependent response relationships, and benefit from training on related relationships when forecasting unseen synthetic mechanisms and historical events. These are the behavioral signatures of inductive forecasting defined in this work.

Forecasting ability is therefore not only a matter of matching outcomes. In these studies, forecasts change systematically with evidence relevance, contextual relationships, and training objectives. The gains are strongest when the relevant relationship is recoverable from the task, and lower forecast error need not imply recovery of the hidden mechanism.

The next test is whether these behaviors persist when cues are unreliable and the relevant variables and mechanisms fall outside the training family.

AI USE STATEMENT

Generative AI systems are central objects of study in this work and produced the forecasts and experimental responses that we evaluate. We also used generative AI tools to generate candidate counterfactual and synthetic materials; refine the conceptual framing, hypotheses, methodology, and experimental design; implement, debug, clean, and reformat code and data; support quantitative analysis and result interpretation; create and edit figures; and draft, edit, structure, and format this manuscript. No formal mathematical theorem or proof is advanced in this work, and translation and qualitative or thematic data analysis were not applicable. AI-generated experimental records were retained for audit and subjected to the programmatic validity checks described in the appendices; AI-assisted code, text, figures, analyses, and claims were reviewed by the authors. The authors take responsibility for the final content of this work, including text, claims, and artifacts produced with the aid of generative AI.

ETHICS STATEMENT

The reported analyses use public market metadata and price histories, model outputs, synthetic worlds, and an unpaid adult review of synthetic materials. The review collected no identifying or sensitive information. Under our institutional guidelines, it does not constitute human-subjects research; consequently, no IRB review was sought. Counterfactual evidence packets were evaluated offline and were not posted to markets or used for trading. We report aggregate model behavior rather than recommend deployment.

REPRODUCIBILITY STATEMENT

Appendices A–C specify the prompts, data-generating processes, sample construction, censoring, estimators, deployment settings, missingness, hyperparameters, frozen baselines, and fail-closed checks. Builders, manifests, checksums, and saved generations support each reported result. Experiment 3’s Qwen3-4B Coin City comparison is complete at the registered round 300 under greedy and five-draw stochastic decoding; the Qwen3-8B and Llama-3.1-8B grids were incomplete when this analysis was finalized. The completed Qwen3-4B arm-by-disclosure factorial is reported as a complementary structural-transfer study in the appendix. The historical test was evaluated only after all three seeds completed, without test feedback during training or model selection.

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A EXPERIMENT 1: PROSPECTIVE FORECAST UPDATING

Experiment 1 measures whether forecast revisions follow evidence direction and whether directional packets produce larger changes than orthogonal packets. All reported counts refer to parseable records in the frozen consistency report unless stated otherwise.

The main comparison reports the nine deployments with scale-valid archived revisions. This appendix retains Qwen 2.5-7B for directional-compliance auditing and documents why its magnitude-based statistics are excluded.

A.1 STUDY DESIGN, SAMPLE, AND AGENT ROSTER

Design at a glance. Experiment 1 crosses ten forecasting agents with a frozen set of unresolved binary Polymarket questions. The study has four stages: (1) freeze the market sample; (2) obtain one or more initial forecasts from each agent; (3) generate nine synthetic evidence packets for each of the 100 core markets (three pro- H_1 , three anti- H_1 , and three causally orthogonal); and (4) ask the same agent to revise each parseable initial forecast after receiving one packet at a time. The unit of analysis is an *agent* $\times$ *market* $\times$ *initial run* $\times$ *counterfactual packet* update. Requested repeat counts were five initial runs for frontier agents and three for local agents; realized valid counts are uneven because of provider failures, parsing failures, and targeted top-ups. Section A.3 specifies how those records are normalized and aggregated.

A.1.1 SAMPLING FRAME AND MARKET SELECTION

The primary sampling frame is a Polymarket API snapshot taken on June 9, 2026. Selection was deterministic given that snapshot. We retained markets that were binary, active and unresolved, scheduled to resolve 14–180 days after sampling, priced strictly between 0.10 and 0.90 on the YES contract, and supported by at least \$1,000 in USD volume. A keyword-based relevance filter retained politics, government, elections, economics, technology, and geopolitics while excluding sports, cryptocurrency, entertainment, and other out-of-scope topics. When several contracts belonged to the same event, we retained the highest-volume contract.

The recorded filter funnel makes the sampling loss explicit. The snapshot contained 60,989 binary records; 37,386 were active, 8,639 passed the topic filter, 3,570 passed the resolution-window filter, 1,403 passed the price filter, 839 passed the volume filter, and event-level deduplication left 436

candidates. A diversity pass selected the 100-market core using a cosine-similarity threshold of 0.25 and a maximum of 16 markets per topical bucket. The frozen core contains 16 global-election, 16 other, 15 Iran-nuclear, 12 U.S.-federal-policy, 10 AI/technology, 9 U.S. state-election, 7 Russia–Ukraine, 5 Brazil, 5 U.S.-midterm, 3 macroeconomic, and 2 China–Taiwan markets.

Counterfactual generation and Stage 3 updating use this 100-market core for every model. Llama 3.1-70B has 99 analyzable markets because one market has no parseable initial forecast.

Table 2: Representative markets from the frozen 100-market core. Snapshot mid is the stored YES midpoint on June 9, 2026, not an eventual outcome.

Question	Topic	Snapshot mid
Will Renan Santos win the 2026 Brazilian presidential election?	Brazil	0.17
US–Iran nuclear deal by June 30?	Middle East	0.20
US × Iran permanent peace deal by July 31, 2026?	Iran	0.32
Will a new Gemini flagship be released by June 30, 2026?	AI/technology	0.89

A.1.2 AGENT ROSTER AND EXECUTION MODES

The roster comprises three hosted frontier systems and seven open-weight local systems (Table 3). The hosted models ran through Azure AI Foundry and used its native search-supported Responses workflow (Microsoft, 2026). Local models were served by Ollama (Ollama, 2026) on cluster GPUs; their research turn was supplied by Tavily search. All agents received the same substantive three-step contract: build an event model, gather current evidence, and emit a structured forecast. Provider-specific search orchestration and model runtimes were not identical.

Table 3: Experiment 1 agent roster and requested initial-forecast repeats. Requested k is the collection target, not an analysis denominator; exact realized coverage appears in Table 6.

Model	Runtime	Research search	Nominal params	Requested k
GPT-5.4	Azure Foundry	Foundry native	—	5
Claude Opus 4.8	Azure Foundry	Foundry native	—	5
DeepSeek V4-Pro	Azure Foundry	Foundry native	—	5
Qwen 2.5-7B	Ollama	Tavily	7B	3
Qwen 2.5-14B	Ollama	Tavily	14B	3
Qwen 2.5-32B	Ollama	Tavily	32B	3
Qwen 2.5-72B	Ollama	Tavily	72B	3
Llama 3.1-8B	Ollama	Tavily	8B	3
Llama 3.1-70B	Ollama	Tavily	70B	3
Llama 3.3-70B	Ollama	Tavily	70B	3

For local models, reasoning calls used the frozen configuration temperature of 0.1 and the final JSON-formatting call used temperature 0.0. Search queries, tool responses, token counts, parse errors, and provider errors were retained with each run. Some smaller models returned probabilities on a 0–100 rather than 0–1 scale. The initial-forecast parser converted those values, but the frozen Stage 3 data retain a mixed-scale Qwen 2.5-7B subset documented in Section A.3.1.

A.2 FORECAST, INTERVENTION, AND VALIDATION PROTOCOL

A.2.1 INITIAL FORECAST ELICITATION

Design rationale. The elicitation asks each agent to restate the resolution criterion, construct explicit YES and NO hypotheses, identify a status-quo trajectory, research current evidence, and map its final YES probability to the posterior assigned to H_1 . Question restatement and status-quo anchoring follow prior forecasting-prompt evidence (Halawi et al., 2024; Wilson, 2025).

Executed three-turn sequence. Every frozen initial forecast followed the same substantive sequence, with provider-specific tool plumbing:

1. **Event model.** The agent received the question, full resolution description, days to resolution, and category. It restated what would count as YES versus NO; identified actors, mechanisms, and latent variables; formed H_1 (YES) and H_2 (NO); and described the status-quo outcome.
2. **Current-evidence research.** The agent gathered evidence for and against each hypothesis, checked whether the required chain of events was plausible within the remaining time, and revised its assessment. Frontier systems used Azure Foundry’s native search workflow; local systems generated queries that were executed through Tavily and received the returned evidence in context.
3. **Structured forecast.** A final call converted the accumulated analysis to the common JSON contract below. A run entered downstream processing only when this object was parseable.

The research turn included this calibration instruction:

Given the days remaining, what chain of events is required for H_1 , and is that pace of change plausible? Give extra weight to the status quo; depart substantially only when concrete directional evidence warrants it.

The common output contract was:

```
{
  "event_model": {
    "key_actors": [str],
    "key_mechanisms": [str],
    "latent_variables": [str]
  },
  "hypotheses": [
    {"id": "H1", "description": str,
     "prior_probability": float, "posterior_probability": float,
     "supporting_evidence": [str], "contradicting_evidence": [str]},
    {"id": "H2", ...}
  ],
  "hypothesis_to_forecast_mapping": str,
  "yes_prob": float,
  "confidence": "low"|"medium"|"high",
  "rationale": str,
  "evidence_sources": [str]
}
```

The contract makes the intended invariant explicit: $\hat{p} = P(H_1)$. The ICS metric tests whether the returned fields obey that invariant after an intervention. The substantive prompt contract was common, but prompts were not byte-identical across runtimes because the hosted and local search interfaces require different orchestration.

A.2.2 COUNTERFACTUAL PACKET CONSTRUCTION AND AUDIT

For each of the 100 core markets, GPT-5.4 generated nine synthetic evidence packets: three pro- H_1 , three anti- H_1 , and three orthogonal. Each generation call received the market question, resolution description, and a canonical parseable initial event model and forecast. It produced:

- **evidence_headline:** one sentence that states the new development;
- **evidence_text:** a 2–4 sentence synthetic wire-style item;
- **mechanism_targeted:** the causal pathway by which the item should affect, or remain irrelevant to, resolution;
- **direction:** pro- H_1 , anti- H_1 , or orthogonal;
- **slot_type:** one of the six roles in Table 4; and
- **plausibility_rating:** the generator’s 1–5 self-rating.

Within each direction, sequential exclusion required a new packet to target a different actor and mechanism from earlier packets. Search was disabled so the packets remained controlled synthetic premises rather than paraphrases of live reporting.

Table 4: Counterfactual slot taxonomy. Each market receives one packet of every direction–slot combination shown, for nine packets total.

Slot type	Direction	Role in the intervention
DATA	pro, anti	A measurement, price signal, statistic, or poll that shifts the likelihood of H_1 .
DECISION	pro, anti	An actor decision, official statement, policy announcement, commitment, or refusal bearing on H_1 .
STRUCTURAL	pro, anti	A technical, logistical, legal, or contextual change that makes H_1 more or less likely.
PROCEDURAL	orthogonal	A same-domain administrative or process update without a direct causal path to H_1 versus H_2 .
PERSONNEL	orthogonal	A staffing or leadership change involving a relevant actor but an issue unrelated to H_1 versus H_2 .
BACKDROP	orthogonal	Contextual economic, logistical, or social information that sets the scene without shifting the outcome odds.

Packet-level integrity checks. The frozen packet file contains 900/900 parseable records: 100 markets multiplied by nine packets, with exactly 300 pro- H_1 , 300 anti- H_1 , and 300 orthogonal records. Every packet has a nonempty headline, and the logs contain zero search calls. All 900 generator-assigned plausibility ratings equal 4/5. That field documents the generator’s compliance claim but provides neither variation nor independent validation; the separate human materials review is reported below. These checks establish structural completeness, not that directional strength or difficulty is perfectly matched across packets.

As an illustration, for the Hormuz market the pro- H_1 headline *Oman says Iran agrees daily protected windows for commercial Hormuz transits* elicited the largest single update reported in the viewer: Claude Opus 4.8 revised from 10% to 45% ($\Delta\hat{p} = +35$ pp). The orthogonal IMO incident-reporting item produced negligible movement.

A.2.3 STAGE 3 UPDATE AND PAIRING

Each packet was presented separately with the agent’s own parseable initial structured forecast. The agent was instructed to accept that single synthetic development as true for the exercise, identify whether it was already implicit in its background view, and return the same structured forecast with revised H_1 , H_2 , and YES probabilities plus a one-sentence `novelty_assessment`. Search was disabled, and agents were instructed to treat the intervention as true without online verification.

For an agent–market cell with k parseable initial runs, the intended crossed design contains $9k$ update records. Each update is conditionally independent in prompt context: it contains one initial run and one packet, never another packet or another agent’s answer. The novelty prompt was intended to discourage large revisions for background information; novelty itself is not used as a reported endpoint and should not be viewed as a validated correction.

Provider interruptions, parse failures, and subsequent top-ups make realized cells uneven. We therefore report exact valid record and market denominators by model in Table 6 rather than treating requested k or the theoretical $9k$ as observed sample sizes.

A.3 ESTIMANDS, RECORD PROCESSING, AND AGGREGATION

A.3.1 RECORD CONSTRUCTION, DEDUPLICATION, AND MISSINGNESS

The frozen analysis artifact is `expl_prospective/data/results/consistency_report_2026-06-17.json`; tables in this paper use its stored denominators rather than a fresh scan of the evolving collection directories. During report construction, date suffixes are stripped from task identifiers so daily collection files join to the same market. Counterfactuals are deduplicated by `cf_id`; updates are deduplicated by `update_id`, with a parseable record preferred over an error record when both exist. Files are read newest first for the initial-forecast and market-price lookup. Malformed JSON lines are skipped. Missing values are handled per estimand: a record may therefore contribute to ICS but not EHC, or vice versa, depending on which fields parse.

Stage 3 update fields were evaluated on their stored scale in the frozen report. For Qwen 2.5-7B, 207 of 8,636 deduplicated records store `updated.yes_prob` above 1.5; no other model does. Its directional sign metrics remain usable, but its absolute revision magnitudes and sensitivity ratio are scale-contaminated and are omitted from the paper tables.

A.3.2 CONSISTENCY ESTIMANDS

For update record i , let $d_i \in \{+1, -1\}$ denote the intended direction (+1 for pro- H_1 , -1 for anti- H_1). Orthogonal packets do not have an intended sign and are excluded from EHC and HFC. The 3 pp movement threshold excludes cases in which a model effectively leaves the forecast unchanged.

Evidence–Hypothesis Consistency (EHC). Let $\Delta p(H_1)_i = p(H_1)_i^{\text{post}} - \hat{p}_i^{\text{initial}}$. For directional records with $|\Delta p(H_1)_i| \geq 0.03$,

$$\text{EHC}_i = \mathbf{1}[\Delta p(H_1)_i d_i > 0]. \quad (2)$$

EHC asks whether the updated hypothesis field moves in the direction implied by the packet; it does not score the size of that movement.

Hypothesis–Forecast Consistency (HFC). Let $\Delta \hat{p}_i = \hat{p}_i^{\text{post}} - \hat{p}_i^{\text{initial}}$. For directional records with $|\Delta \hat{p}_i| \geq 0.03$,

$$\text{HFC}_i = \mathbf{1}[\Delta \hat{p}_i d_i > 0]. \quad (3)$$

HFC asks the same directional question of the headline YES probability. Because the output contract requires this probability to equal $P(H_1)$, we retain HFC in the appendix as a compliance diagnostic rather than as a separate main-text result.

Internal Coherence Score (ICS). ICS is defined whenever both updated fields parse and is unconditional on movement:

$$\text{ICS}_i = \mathbf{1}[|\hat{p}_i^{\text{post}} - p(H_1)_i^{\text{post}}| < 0.02]. \quad (4)$$

The inequality is strict: a discrepancy of exactly 2 pp does not pass.

Directional sensitivity. Packet selectivity is summarized by

$$\text{Sensitivity} = \frac{|\Delta \hat{p}|_{\text{pro+anti}}}{|\Delta \hat{p}|_{\text{orthogonal}}}. \quad (5)$$

A value near one indicates comparable movement for directional and orthogonal packets; larger values indicate preferential response to directional packets. This ratio is descriptive. It cancels a uniform multiplicative scale within a model, but does not repair mixed scales across records, hence the Qwen 2.5-7B caveat above.

A.3.3 AGGREGATION AND UNCERTAINTY

For EHC, HFC, and ICS, the evaluator first pools eligible records within each model–market cell to obtain a market-level rate. The reported model score is the unweighted mean of those market rates, so a market with more successful repeat runs does not receive greater weight. Reported standard errors are the sample standard deviation of market-level rates divided by the square root of the number of markets.

Revision magnitudes and sensitivity use the deduplicated update records pooled within model and direction. Their standard errors are record-level summaries and do not model dependence among repeated initial runs or the nine packets from one market; they should be read descriptively.

A.3.4 OUTPUT COMPLIANCE AND THRESHOLD ROBUSTNESS

The two output fields are structurally coupled. The final-output schema explicitly instructs the agent that $\hat{p} = P(H_1)$, and the Stage 3 prompt requests the same structure after updating. EHC and HFC are therefore not independent tests of an inferred hypothesis-to-forecast link. ICS is a schema-adherence diagnostic, while close EHC–HFC agreement reflects a mixture of directional updating

and compliance with the prompted equality. Among the 26,010 relevant records with both updated fields present, 25,984 (99.90%) satisfy the equality exactly after scale normalisation and 25,987 (99.91%) satisfy it within the 2-pp ICS tolerance. We therefore do not treat EHC–HFC parity as independent evidence for a latent representation or for spontaneous propagation between separately elicited quantities.

Unconditional estimand. For threshold t , Table 5 retains a valid directional update when $|\Delta| \geq t$ and reports the fraction for which $d_i \Delta_i > 0$. At $t = 0$, every valid pro- and anti- H_1 update enters the denominator and a zero change is scored incorrect. We first compute correctness within market and then average markets equally, matching the primary analysis. The *Below 3 pp* column is the record-weighted fraction and count excluded by the published 3-pp conditional metric. The table’s n is the complete pre-threshold denominator: records require a parseable initial probability and the applicable updated field, but are not selected on the direction or magnitude of movement.

Table 5: Unconditional directional correctness and robustness to 0-, 1-, 3-, and 5-percentage-point movement thresholds. The 0-pp column is unconditional over all valid directional updates and scores zero movement as incorrect; positive-threshold columns condition on the stated magnitude. *Below 3 pp* gives the percentage and count omitted from the corresponding published 3-pp EHC or HFC denominator. Rates are macro-averaged across markets.

Model	Metric	n valid	Directional correctness				
			0 pp	1 pp	3 pp	5 pp	Below 3 pp
Claude Opus 4.8	EHC	2,119	98.2%	98.6%	99.0%	99.5%	18.1% (383)
	HFC	2,119	98.3%	98.6%	99.0%	99.5%	18.1% (383)
GPT-5.4	EHC	2,678	98.4%	98.4%	98.7%	98.8%	6.9% (185)
	HFC	2,684	98.4%	98.4%	98.7%	98.8%	6.9% (185)
DeepSeek V4-Pro	EHC	2,929	97.3%	97.6%	97.7%	98.0%	12.7% (371)
	HFC	2,929	97.3%	97.6%	97.7%	98.0%	12.6% (370)
Llama 3.3-70B	EHC	1,788	96.4%	96.4%	96.5%	97.4%	0.7% (12)
	HFC	1,788	96.4%	96.4%	96.5%	97.4%	0.7% (12)
Llama 3.1-70B	EHC	1,659	96.1%	96.1%	96.1%	96.7%	0.6% (10)
	HFC	1,659	96.1%	96.1%	96.1%	96.7%	0.6% (10)
Qwen 2.5-72B	EHC	1,795	96.3%	96.3%	96.3%	96.4%	0.8% (15)
	HFC	1,795	96.3%	96.3%	96.3%	96.4%	0.8% (15)
Qwen 2.5-32B	EHC	1,795	96.4%	96.4%	96.4%	96.5%	1.9% (34)
	HFC	1,795	96.4%	96.4%	96.4%	96.5%	1.9% (34)
Qwen 2.5-14B	EHC	1,800	93.1%	94.4%	94.3%	94.3%	2.9% (52)
	HFC	1,800	93.1%	94.4%	94.3%	94.3%	2.9% (52)
Qwen 2.5-7B	EHC	5,759	90.2%	90.2%	90.6%	90.8%	2.2% (129)
	HFC	5,759	89.9%	90.2%	90.5%	90.8%	2.4% (140)
Llama 3.1-8B	EHC	3,601	88.1%	88.7%	88.7%	88.9%	1.7% (62)
	HFC	3,604	88.4%	88.7%	88.7%	88.9%	1.4% (52)

Result. The threshold sweep preserves the qualitative conclusion but narrows its wording. Frontier-model unconditional correctness is 97.3–98.4%; at the published 3-pp threshold it is 97.7–99.0%. The 3-pp rule excludes 6.9% of GPT-5.4, 12.6–12.7% of DeepSeek V4-Pro, and 18.1% of Claude Opus 4.8 directional updates. Thus, the near-1.0 EHC/HFC values do not result from discarding a majority of cases, and the conclusion is stable across 0, 1, 3, and 5 pp. The precise statement is nonetheless *99% directionally correct among updates of at least 3 pp*, not *99% coherent updating*. Small local models have fewer sub-3-pp updates (0.6–2.9%) but lower unconditional correctness (88.1–96.4%), so their errors are not hidden primarily by the movement filter.

Normalisation and reproducibility. Before recomputing deltas, the analysis independently normalises the initial headline probability, updated headline probability, and updated H_1 posterior

from the documented $[0, 1]$ or $[0, 100]$ formats. This matters for 166 relevant Llama 3.1-8B records in which H_1 was emitted as a percentage while `yes_prob` was emitted as a proportion. The complete eligible record and market counts at every threshold, invalid-field counts, market-level standard errors, input-file SHA-256 hashes, and full-precision estimates are in `expl_prospective/data/results/threshold_robustness.json`. The table is regenerated by `expl_prospective/agent/evaluate_threshold_robustness.py`; neither script nor table uses outcomes or post-hoc item filtering.

A.4 EXTENDED RESULTS

A.4.1 ANALYSIS COVERAGE

Table 6 gives the exact Stage 3 record and market counts. All models use the same 100-market core; Llama 3.1-70B has 99 analyzable markets because one market has no parseable initial forecast.

Table 6: Valid Stage 3 update records and represented core markets by agent.

Model	Type	Records	Markets
Claude Opus 4.8	Frontier	3,125	100
GPT-5.4	Frontier	3,913	100
DeepSeek V4-Pro	Frontier	4,395	100
Llama 3.3-70B	Local 70B	2,682	100
Llama 3.1-70B	Local 70B	2,490	99
Qwen 2.5-72B	Local 72B	2,692	100
Qwen 2.5-32B	Local 32B	2,693	100
Qwen 2.5-14B	Local 14B	2,700	100
Qwen 2.5-7B	Local 7B	8,636	100
Llama 3.1-8B	Local 8B	5,404	100
Total		38,730	

A.4.2 REVISION MAGNITUDES AND PACKET SELECTIVITY

Table 7 reports mean absolute forecast revision $|\Delta\hat{p}|$ by packet direction. For the nine models whose updates are consistently stored on 0–1, frontier systems show a clear separation between directional and orthogonal packets, while local systems show weaker separation.

Qwen 2.5-7B has mixed 0–1 and 0–100 stored updates (Section A.3.1); we therefore omit its absolute magnitudes and sensitivity ratio rather than present them as comparable estimates.

Table 7: Mean absolute forecast revision by counterfactual direction. Sensitivity is mean directional divided by mean orthogonal movement; n is the valid Stage 3 record count. Values are percentage points for models with consistently stored 0–1 updates.

Model	Type	pro- H_1	anti- H_1	orthogonal	Sensitivity	n
Claude Opus 4.8	Frontier	12.7 pp	9.6 pp	0.5 pp	24.0×	3,125
DeepSeek V4-Pro	Frontier	13.9 pp	11.4 pp	1.4 pp	8.9×	4,395
GPT-5.4	Frontier	10.6 pp	8.5 pp	1.2 pp	8.2×	3,913
Llama 3.3-70B	Local 70B	14.8 pp	14.3 pp	2.3 pp	6.4×	2,682
Llama 3.1-70B	Local 70B	14.8 pp	15.5 pp	3.3 pp	4.6×	2,490
Qwen 2.5-72B	Local 72B	15.0 pp	10.2 pp	3.0 pp	4.2×	2,692
Qwen 2.5-32B	Local 32B	15.0 pp	10.3 pp	4.0 pp	3.2×	2,693
Qwen 2.5-14B	Local 14B	14.8 pp	8.6 pp	4.1 pp	2.9×	2,700
Qwen 2.5-7B	Local 7B	—	—	—	—	8,636
Llama 3.1-8B	Local 8B	11.1 pp	12.0 pp	5.9 pp	1.9×	5,404

Qwen 2.5-7B magnitude cells are omitted because 207 deduplicated updates are stored above 1.5, mixing probability scales within the row. Table 7 establishes the magnitude contrast; Figure 8 then places directional consistency and packet selectivity on a common visual scale.

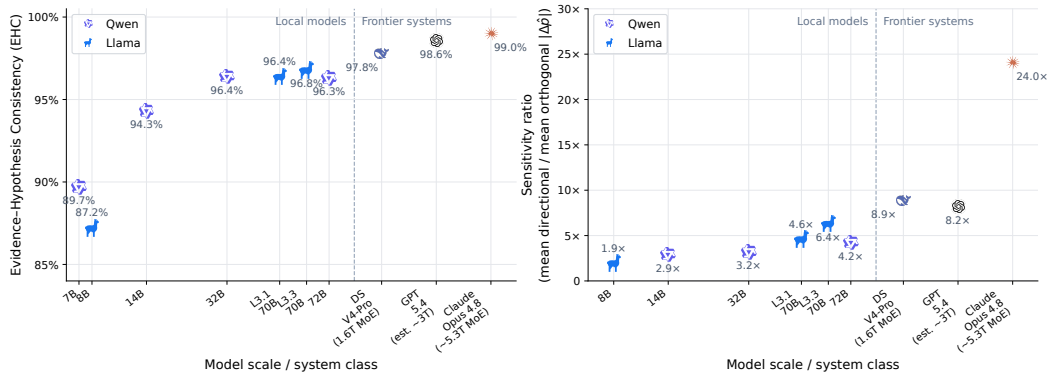


Figure 8: **Directional consistency and packet selectivity generally rise with model scale.** *Left:* EHC rises from 87–90% for 7–8B local models to roughly 98–99% for frontier systems. *Right:* sensitivity reaches 1.9–6.4× for local models and 8.2–24.0× for frontier systems; Qwen 2.5-7B is omitted from this panel because its archived absolute revisions mix 0–1 and 0–100 scales and therefore do not yield a comparable ratio.

A.5 STUDY LIMITATIONS

A.5.1 SAMPLING AND OUTCOME SCOPE

The sample is a filtered slice of liquid, medium-horizon Polymarket questions, not a probability sample of forecasting tasks. Topic exclusions, the 0.10–0.90 price window, the 14–180 day horizon, event deduplication, and the diversity cap shape its difficulty distribution. Experiment 1 evaluates conditional updating on unresolved questions; Experiment 4 separately evaluates accuracy on resolved historical markets.

A.5.2 COUNTERFACTUAL CONSTRUCT VALIDITY

The intervention audit establishes structural completeness, but not equal difficulty. A single generator created all packets from a canonical initial event model, and its automated plausibility score was uniformly 4/5. Direct headlines can also permit a semantic sign classifier to pass without following the intended causal path. The reported result therefore establishes direction- and relevance-sensitive updating under semantically direct packets; it does not establish mechanism-conditioned causal composition by itself.

A.5.3 HUMAN MATERIALS VALIDATION

We froze an initial seven-reviewer human materials protocol before registered collection. It samples one report from each of 18 distinct markets, balanced across six pro-YES, six anti-YES, and six orthogonal packets. Two later roster extensions, made after initial responses were inspected, added independently ordered assignments for R8 and R9 without changing the items or any earlier assignment; the resulting nine-reviewer analysis is a sensitivity extension. Each reviewer rates all 18 items. Before the synthetic report, every item shows the market question, resolution criteria, and matching summary from the frozen June 10 initial forecast. The consent screen and every item instruct reviewers to treat June 10, 2026 as the current date, use no later information, and not search the web or consult an AI assistant. Intended directions, source IDs, generator metadata, and model responses remain hidden.

Reviewers record the report’s conditional direction, confidence, clarity, real-world plausibility, and usability as an explicit hypothetical premise. The frozen seven-reviewer item gate allowed one dissent: at least six reviewers had to select the registered direction and mark the premise usable, median confidence and clarity had to be at least 4/5, and median plausibility at least 3/5. The nine-reviewer extension uses the corresponding eight-of-nine rule. The packet-level gate requires at least 15 of 18 items to pass and Fleiss’ $\kappa \geq .70$ for conditional direction. Ambiguous judgments remain in every denominator.

A pilot review conducted before registered collection removed three non-gating questions—semantic directness, source verification, and optional free text—and added the frozen June 10 background summary to every item. Practice responses are stored separately and cannot enter the registered analysis.

In the frozen August 23 UTC analysis, eight reviewers had completed all 18 items. The post-collection response-pattern rule described below excludes R2, leaving seven completed, quality-eligible reviewers (R1, R3–R7, and R9). R8 had consented but submitted no ratings, so the final nine-reviewer gate is not yet available. Table 8 reports the completed-reviewer analysis. Exact-direction agreement is 96/126 (76.2%); every included reviewer exceeds a conservative 1/3 directional-guessing baseline in a one-sided exact binomial comparison (all unadjusted $p < .05$). We use 1/3 rather than the form’s four-option 1/4 baseline because the registered answer takes one of three non-ambiguous directions. The task requires reviewers to integrate unfamiliar market-specific background with a stipulated synthetic report and distinguish directional, null, and ambiguous updates.

The completed-reviewer analysis does not meet the gate. Applying the full gate—including premise usability—to the seven quality-eligible completed reviewers retains 5 of 18 items. Considering direction alone, eight items draw at most one dissent and five are unanimous. Fleiss’ κ for conditional direction is .480 over the four response options (.530 if *ambiguous* is folded into the null category), and mean pairwise Cohen’s κ is .484 (range .092 to .838). These values remain below the packet-level thresholds of 15 retained items and $\kappa \geq .70$. Because the roster is incomplete, this analysis is not a final gate decision and does not certify individual packets. The majority reviewer label matches the registered direction on 16 of 18 items ($\kappa = .842$), and per-class exact agreement is 29/42 for pro- H_1 , 33/42 for anti- H_1 , and 34/42 for orthogonal packets.

Table 8: **Stage 3 human materials judgments for seven completed reviewers.** Direction is exact agreement with the registered conditional direction and κ is Cohen’s κ against that key over the four response options. Confidence, clarity, and plausibility (C/C/P) are reviewer-level medians on 1–5 scales. The pooled row is descriptive because reviewers rated the same items; the majority row takes the modal label per item, resolving ties to *ambiguous*.

Reviewer	Direction agreement	κ vs. key	p vs. 1/3	Premise usable	C/C/P
R1	12/18 (66.7%)	.54	.0039	10/18	4/4/4
R3	12/18 (66.7%)	.51	.0039	13/18	4/4/3
R4	18/18 (100.0%)	1.00	< .0001	18/18	5/5/5
R5	10/18 (55.6%)	.38	.0433	18/18	4/4/4
R6	13/18 (72.2%)	.61	.0009	12/18	4/4/5
R7	15/18 (83.3%)	.76	< .0001	17/18	5/4/3
R9	16/18 (88.9%)	.84	< .0001	18/18	5/5/5
Pooled	96/126 (76.2%)	—	—	106/126	—
Majority label	16/18 (88.9%)	.84	—	—	—

Table 9: **Item-level Stage 3 materials audit for seven completed reviewers.** Key is the registered conditional direction; Majority is the modal panel response, with ties assigned to *ambiguous*. Agree and Usable give counts out of seven; C/L/P gives median direction confidence, clarity, and plausibility on 1–5 scales. The seven-reviewer diagnostic requires at least 6/7 registered-direction judgments, at least 6/7 usable-premise judgments, median confidence and clarity of at least 4, and median plausibility of at least 3. Failure codes identify the unmet criterion: D (direction), C (confidence), L (clarity), P (plausibility), or U (premise usability). Item numbers follow the frozen public packet order.

Question	Key	Majority	Agree	C/L/P	Usable	Gate
1. Hormuz traffic normal by July 31	More	More	7/7	5/5/5	7/7	Pass
2. Iran reaches World Cup knockouts	Less	Less	5/7	5/5/5	6/7	Fail (D)
3. Israel strikes Yemen by June 30	No effect	No effect	7/7	5/4/4	5/7	Fail (U)
4. Netanyahu exits election by July 31	More	More	5/7	5/5/4	7/7	Fail (D)
5. Anthropic valuation reaches \$1.1T	Less	Less	6/7	4/4/5	7/7	Pass
6. Oladokun starts Chiefs Week 1	No effect	No effect	6/7	4/4/3	6/7	Pass
7. S&P 500 reaches 7,700 in June	More	Ambig.	3/7	4/4/5	6/7	Fail (D)
8. Trump unfreezes Iranian assets	Less	Less	5/7	4/4/4	6/7	Fail (D)
9. May durable-goods orders: 0–2%	No effect	No effect	4/7	3/3/3	5/7	Fail (D,C,L,U)
10. China Q2 growth: 4.6–4.9%	More	Ambig.	2/7	3/4/4	4/7	Fail (D,C,U)
11. Eisenkot joins Bennett–Lapid alliance	Less	Less	5/7	5/5/5	6/7	Fail (D)
12. Russia enters Orikhiv by July 31	No effect	No effect	7/7	5/4/4	5/7	Fail (U)
13. Romanian PM is independent/technocrat	More	More	5/7	4/4/4	7/7	Fail (D)
14. Russia captures Kostyantynivka	Less	Less	5/7	3/5/5	6/7	Fail (D,C)
15. At least four June ChatGPT outages	No effect	No effect	4/7	3/4/4	6/7	Fail (D,C)
16. El-Sayed wins Michigan primary	More	More	7/7	5/5/5	7/7	Pass
17. Little is MN-02 Democratic nominee	Less	Less	7/7	5/5/4	6/7	Pass
18. OpenAI IPO close: \$1.25–1.5T	No effect	No effect	6/7	4/3/4	4/7	Fail (L,U)

After inspecting the first four completed submissions, we excluded one completed reviewer (R2) for task misunderstanding: the reviewer selected *more likely* on 17 of 18 balanced items, including all six registered negative items and five of six registered null items. This is a disclosed post-collection deviation from the frozen no-exclusion rule. We applied the resulting quality rule uniformly: exclude any completed reviewer who selects one direction on at least 17 of 18 items. Raw responses are retained, and no item is removed or relabeled. As a sensitivity tally, retaining R2 gives 103/144 (71.5%) exact agreement across all eight completed reviewers.

A post-task debrief occurred after R2 had seen their score and response pattern. R2 reported interpreting the task as asking whether reports supported YES, difficulty with English and market-specific terms, use of a dictionary and an AI assistant despite the instruction, and reduced attention. We accepted no re-rating and treat the debrief as context; the exclusion uses only the uniform response-pattern rule.

The task requests no identifying or sensitive information. Under the authors’ institutional guidelines, this materials-review activity does not constitute human-subjects research; consequently, no IRB review was sought.

Figure 9 records the reviewer-facing workflow. The capture script at `expl_prospective/stage3_materials_annotation/capture_review_screenshots.py` loads the frozen item packet against an empty temporary annotation database. Later reviewer assignments changed access codes and item order only.

Orthogonal packets are a useful causal-irrelevance control, but are not equivalent to receiving no new text. A literal no-information update condition would separate generic prompt-induced revision from response to topically adjacent content. The very small frontier orthogonal revisions (0.5–1.4 pp) are reassuring but do not replace that control.

A.5.4 PROMPT, RETRIEVAL, AND RUNTIME CONFOUNDING

Agents shared a substantive task contract, not byte-identical execution. Hosted models used Azure Foundry search, while local models used Tavily results in an Ollama workflow; temperatures and provider behavior also differ. Frontier–local gaps therefore combine model capability with retrieval and runtime differences.

(a) Information, temporal frame, and consent.

(b) A complete item with June 10 context, synthetic report, and five judgments.

1402

1403

A.5.5 SCALE, MISSINGNESS, AND DEPENDENCE

Realized repeat counts are uneven because failures and top-ups differ by provider. Equal-weight market aggregation prevents high-coverage markets from dominating EHC, HFC, and ICS, but record-level revision summaries still treat repeated runs and packets as independent. Clustered or hierarchical uncertainty would be preferable. The mixed-scale Qwen 2.5-7B updates described in Section A.3.1 additionally invalidate a clean magnitude comparison for that row; a future frozen rerun should normalize every structured field before deltas are written.

A.5.6 PREMISE ACCEPTANCE AT STAGE 3

Search is intentionally unavailable during updating, and agents are told to accept the synthetic headline as true. This isolates conditional updating but does not test source verification or epistemic refusal. A search-enabled condition might appropriately decline to update on an unverifiable report, lowering EHC/HFC without implying less inductive reasoning. The human materials review above uses the same conditional premise instruction and therefore assesses comprehension and usability, not source credibility. Both premise-acceptance and source-verification conditions are needed to distinguish these behaviors.

A.6 REPRODUCIBILITY AND ARTIFACT MAP

The paper treats the June 17 report as the frozen numerical authority. The relevant artifacts are:

- **Sampling:** `expl_prospective/fetch_markets/select_markets.py` and the June 9 selection/diversity JSON and manifests in `expl_prospective/data/selected_markets/`.
- **Initial elicitation:** `expl_prospective/agent/forecast_agent.py` for hosted runs and `expl_prospective/agent/run_local_forecast.py` for Ollama runs, with records and manifests in `expl_prospective/data/initial_forecasts/`.
- **Interventions:** `expl_prospective/agent/build_counterfactuals.py` and the frozen `expl_prospective/data/counterfactuals/counterfactuals_2026-06-10.jsonl`.
- **Updates:** hosted and local update runners under `expl_prospective/agent/`, with update records in `expl_prospective/data/updated_forecasts/`.
- **Evaluation:** `expl_prospective/agent/evaluate_consistency.py` and `expl_prospective/data/results/consistency_report_2026-06-17.json`.
- **Human materials review:** the frozen protocol, blinded packet, browser application, tests, and analysis script in `expl_prospective/stage3_materials_annotation/`.
- **Figures:** `paper/generate_figures.py` writes the direction-selective panel; `paper/generate_paper_figures.py` writes the combined EHC/sensitivity panel.

The model-update tables are defined by the June 17 frozen JSON. The mutable collection directories contain later records, so evaluating their present contents would define a different dataset. Bit-for-bit reconstruction begins from the frozen JSON; a full pipeline reconstruction uses only the manifests and records included in that freeze.

B EXPERIMENT 2: CONTEXT-GUIDED LATENT-RESPONSE INFERENCE IN COIN CITY

Experiment 2 tests whether models use context and sparse observations to infer a response relationship. All six deployments reuse the same 250 seeded episodes, so model and prompt-arm differences are paired rather than comparisons between independently sampled tests.

B.1 FIGURE 1 CARNIVAL-COIN FORECASTS

The model markers in Figure 1 are empirical forecasts, not illustrative placements. We queried GPT-5.4, Claude Opus 4.8, and DeepSeek V4-Pro eight times each (24 calls total) at temperature

0.7 and a 600-token output ceiling. All 24 responses parsed. The figure places each model’s logo at its eight-sample mean; the green $\approx .68$ badge rounds the pooled mean of .679.

The exact user prompt was:

You are at a carnival game. A coin you have never seen before is flipped twice in front of you, and it comes up heads both times. These two flips are everything you know about this coin.

What is the probability that the next flip comes up heads?

Think about it intuitively, the way you actually would in the moment -- do NOT use any formula, rule, or calculation, and do not work out any numbers in your reasoning. Just reason in plain words about what you’d expect and why. Then END your reply with your gut answer as a JSON object on its own line:

```
{"rationale": "one short sentence, no math",
  "p_heads": <number between 0 and 1>}
```

Table 10: Forecasts underlying the three model markers in Figure 1. Each row summarizes eight independent temperature-0.7 calls.

Model	Mean	Median	Range
GPT-5.4	.644	.670	[.60,.67]
Claude Opus 4.8	.681	.675	[.65,.75]
DeepSeek V4-Pro	.712	.750	[.60,.75]
Pooled ($n = 24$)	.679	.670	[.60,.75]

The prompt, temperature, output limit, parsed probabilities, rationales, and raw completions are stored in `exp2_simulated_worlds/biased_news/data/coin_probe/coin_probe_2026-06-30.jsonl`; the collection and parsing code is `exp2_simulated_worlds/biased_news/eval/run_coin_probe.py`. The prompt names neither a fair nor a rigged coin. It asks for qualitative reasoning to avoid collapsing the probe into a requested formula, while requiring a numeric forecast for the plotted quantity.

B.2 DESIGN, DATA-GENERATING PROCESS, AND FREEZE

B.2.1 DESIGN AT A GLANCE

The frozen design identifier is `coin_city_stable_relationship_claude_n250_v4`. It contains 250 independent episodes generated from seeds 930000–930249, five target-city prefixes $k \in \{0, 1, 2, 3, 4\}$, and three primary prompt arms frozen before model calls. The misleading-context and arbitrary-symbol arms are later, additive controls described in Section B.4. Every model received the same episode records, queries, and arm-specific prompt text.

B.2.2 BALANCED LATENT REGIMES

Each episode crosses two binary factors: whether City A or City B is the strongly responsive reference city, and whether target City C is strongly or weakly responsive. The closest-balanced allocation contains 63 A-strong/C-strong, 62 A-strong/C-weak, 62 B-strong/C-strong, and 63 B-strong/C-weak episodes. Within an episode, each city receives an independently sampled stable slope

$$g_{ec} \sim \begin{cases} \mathcal{N}(0.90, 0.03^2), & \text{strong regime,} \\ \mathcal{N}(0.25, 0.03^2), & \text{weak regime.} \end{cases}$$

Thus the cue identifies a response regime but not the realized coefficient.

B.2.3 OBSERVED CASES AND FORECAST QUERY

Every city has four reference cases arranged as two matched pairs. Rows in a pair share a starting poll and news magnitude but have opposite signs; the first sign is random. For reference row r ,

$$p_{ecr}^{\text{end}} = \text{round}_{0.1} \left(p_{ecr}^{\text{start}} + g_{ec} x_{ecr} + \epsilon_{ecr} \right), \quad \epsilon_{ecr} \sim \mathcal{N}(0, 5^2).$$

Starting polls are uniform on $[35, 65]$, and reference-news magnitudes are drawn from $\{6, 7, 8, 9, 10\}$. The query shock is drawn from $\{-10, \dots, -4, 4, \dots, 10\}$. Its scoring target is the latent conditional expectation $p_0 + g_C x$, with no additional query-time observation noise.

The prefix k exposes the first k City C rows in their frozen order. Consequently, $k = 2$ completes the first matched pair and $k = 4$ completes both; $k = 1$ and $k = 3$ deliberately expose partial pairs. City A and B always contribute all four rows in the ABC arms.

Table 11: Frozen Coin City design and evaluation dimensions. Final analytic record counts retain one record per task cell; raw transport-level calls are discussed in Section B.7.

Quantity	Frozen value
Independent episodes	250
Factorial cells	63, 62, 62, 63 across the four A/B-by-C regimes
Target-city prefixes	$k \in \{0, 1, 2, 3, 4\}$
Primary prompt arms	target only; ABC/no C context; ABC/correct C context
Additive control arms	ABC/misleading C context; ABC/episode-randomized arbitrary labels
Evaluated deployments	Claude Opus 4.8; GPT-5.6; DeepSeek-V4-Pro; Kimi K3; Gemini 3.6 Flash; Claude Opus 5
Primary task cells / final records	22,500 / 22,500 (22,497 parseable)
Misleading task cells / final records	7,500 / 7,500 (all parseable)
Symbol-control matched task cells / final records	3,750 / 3,750 (DeepSeek; all parseable)
Strong / weak slope means	0.90 / 0.25 poll points per news point
Slope standard deviation	0.03 within each regime
Reference-case noise	5 poll points
Reference cases per city	4 (two matched pairs)

B.2.4 FREEZE AND PREFLIGHT VALIDATION

The GPT-5.6, DeepSeek-V4-Pro, Kimi K3, Gemini 3.6 Flash, and Claude Opus 5 replications reuse the exact task files and prompt hashes created for the original Claude Opus 4.8 evaluation; no episode was regenerated. Before model calls, the manifest hashed the engine, task files, answer and scoring keys, validator, runner, renderer, and launcher. Automated validation confirmed the intended dimensions and factorial balance, prompt contracts and hashes, correct cue-to-regime mapping, visible matched pairs, and benchmark reconstruction. No model response or performance statistic was used by this preflight validator.

B.3 PROMPT ARMS AND INFORMATION AVAILABLE

B.3.1 SHARED TASK INSTRUCTIONS

All prompts state that rows are separate polling cases rather than a time series, positive news favors the candidate, poll change is end minus start, and within-city responsiveness is stable even though individual polls are noisy. Rows sharing a pair label are described as having the same starting poll and equal-magnitude, opposite-sign news. Every displayed row includes the starting poll, news value, change, and ending poll.

The required answer is a JSON object with exactly a one-sentence `rationale` and a numeric `predicted_poll` between 0 and 100. The prompt asks for no step-by-step derivation.

B.3.2 FIVE INFORMATION CONDITIONS

Target only. The model sees the first k City C cases and is explicitly told that no additional background information is available. This arm measures recovery from target-city evidence alone.

ABC, no City C context. The model also sees four cases from each reference city. The strongly responsive reference city is described as receiving sustained national coverage, whereas the weakly responsive city receives local coverage where campaign stories compete with

local issues. City C receives no background description. This arm tests numeric transfer without a cue that aligns C to either reference regime.

ABC plus correct City C context. The numeric rows and query are identical to the preceding arm. The sole added information is that City C receives national or local coverage, correctly aligned with its latent strong or weak regime. This is the primary cue-transfer condition.

ABC plus misleading City C context. This additive arm is identical to the correct-context prompt except that its City C sentence names the opposite coverage regime. Every number, case row, query, and other instruction is unchanged. This arm tests whether the model follows the semantic cue when it conflicts with the truth.

ABC plus an arbitrary City C label. Every national/local description is replaced by one of two arbitrary labels, KIV or ZOR. The prompt states only that cities sharing a label share a typical response regime and that the labels have no inherent meaning or ordering. Their higher/lower-response mapping reverses on alternating episodes; City A and B’s numerical cases are therefore the only evidence from which the episode-local mapping can be induced before applying City C’s label.

No prompt contains a fitted coefficient, an estimator, the strong or weak numerical means, the case-noise scale, a rule for pooling cities, or any analyst benchmark. The correct-versus-no-context contrast therefore isolates the one City C background sentence while holding all displayed numeric evidence fixed; the correct-versus-misleading comparison isolates the sentence’s regime assignment. The symbol-versus-no-context contrast asks whether the same assignment can be induced without lexical knowledge of national and local coverage.

B.4 MODEL EXECUTION AND RESPONSE PROCESSING

B.4.1 DEPLOYMENT CONFIGURATIONS

Table 12 reports the effective API settings used for production. Temperature refers to the value sent to the provider: GPT-5.6 and Claude Opus 5 omit it even though the local GPT run manifest records a default of zero. No deployment used tools or retrieval.

Table 12: Production deployment configuration and primary-arm parse counts.

Deployment	Interface	Reasoning / sampling	Output ceiling	Parsed
Claude Opus 4.8	Anthropic Messages via Azure	temperature 0	512	3,750
GPT-5.6	OpenAI Responses	low reasoning; temperature omitted	4,096	3,749
DeepSeek-V4-Pro	OpenAI Chat Completions	temperature 0	4,096	3,750
Kimi K3	OpenAI Chat Completions	low reasoning; temperature 0	16,384	3,750
Gemini 3.6 Flash	Gemini OpenAI-compatible Chat	low reasoning; temperature 0	16,384	3,750
Claude Opus 5	Foundry Anthropic Messages	default extended thinking; temperature omitted	4,096	3,748

Claude Opus 5 returns extended thinking before its answer; the runner therefore selects its first text block. Gemini is the sole fast-tier deployment, so the six systems should be read as a replication set, not a tier-matched leaderboard. Authentication material was supplied at runtime and was not written to source, manifests, logs, or response files.

B.4.2 ATTEMPTS, TRANSPORT RETRIES, AND EXCLUDED PILOTS

Every primary model-prompt pair received one application-level attempt. A failed or answerless task was not re-asked. HTTP-client retries were disabled for GPT, DeepSeek, Kimi, and Opus 5. The original Opus 4.8 client allowed up to two transport retries. Gemini’s runner allowed adaptive pre-model transport retries but recorded none in the completed shards. These transport policies do not alter the one-answer-per-task analytic rule.

Before production, 119 Kimi, 15 Gemini, and 17 Opus 5 task-shaped calls were used only to select configurations that reliably returned the required JSON schema. Forecast accuracy was not inspected. The calls are archived separately and are excluded from every denominator and analysis; each production namespace began empty. Production tasks were sharded only for throughput.

B.4.3 RESPONSE PARSING AND RECORD SELECTION

The parser scans the raw reply for JSON objects, retains the last object with an exact `predicted_poll` key whose value converts to a number in $[0, 100]$, truncates the retained rationale to 500 characters, and stores at most 8,000 raw characters. When duplicate task records exist, the analysis keeps the latest parseable response, allowing a valid response to supersede an earlier error. A single offline normalization removed whitespace from JSON keys in one otherwise valid Opus 4.8 reply; this recovered one primary record without another model call. Other production-file repairs are documented in Section B.7.

B.4.4 ADDITIVE MISLEADING-CONTEXT EXECUTION

The misleading arm was generated after completion of the three frozen primary arms. Its builder verifies that every prompt equals its correct-context counterpart under exactly one sentence substitution and that all numeric content is unchanged. The cue is inverted for all 1,250 episode-prefix tasks: 625 strong-city tasks are described as weak and 625 weak-city tasks as strong. Each deployment used the same client and production configuration as in its own correct-context run. The original three-arm hashes remain unchanged; this stress test is additive rather than part of the initial freeze.

B.4.5 ADDITIVE ARBITRARY-SYMBOL EXECUTION

The symbol builder generates 1,250 additional prompts from the same frozen episodes and verifies that their complete sequence of numeric tokens is identical to the correct-context arm. The higher-response label is `KIV` in 125 episodes and `ZOR` in 125; the target regime-by-label cells contain 62–63 episodes each. A prompt validator rejects national/local and strong/weak semantic descriptions. At $k = 0$, City C supplies no numerical cases, so its correct regime can be recovered only by decoding the label from City A and B’s examples.

We executed this control on DeepSeek-V4-Pro. To prevent a provider-resource change from contaminating arm contrasts, we collected all 1,250 symbol tasks and contemporaneously re-collected all 2,500 no-context and semantic-context tasks on the same reachable Azure resource, using the same deployment identifier, client, temperature, and output budget. These records occupy a separate additive namespace; no primary response file or frozen task was overwritten.

B.5 ESTIMANDS, BENCHMARKS, AND STATISTICAL INFERENCE

B.5.1 FORECAST AND LATENT-RESPONSE ESTIMANDS

Forecast error is measured against the latent conditional expectation rather than a newly sampled noisy ending poll:

$$\text{MAE} = \frac{1}{n} \sum_{e=1}^n |\hat{p}_e - (p_{0e} + g_{Ce}x_e)|.$$

For every nonzero query shock, a forecast also identifies an implied slope $\hat{g}_e = (\hat{p}_e - p_{0e})/x_e$. We report its mean absolute error from g_{Ce} and its Pearson correlation with g_{Ce} across episodes. Correlation is needed because a nearly constant prior-mean slope can attain moderate MAE while failing to distinguish strong from weak cities.

Within each deployment and prefix, the three primary-arm metrics use the common intersection of episodes with parseable responses in all three arms. Arm contrasts are paired by episode and prefix. The misleading-arm table uses its final unique parseable record for every task. The arbitrary-symbol analysis uses the corresponding common intersection of the three contemporaneous DeepSeek arms.

B.5.2 ANALYST-ONLY BENCHMARKS

The ABC OLS benchmark fits poll change on news through the origin using every displayed A, B, and available C row. The target-only OLS benchmark uses only displayed City C rows and is undefined at $k = 0$. Both are reconstructed from the visible data by the analyst; neither the estimator nor its output appears in a model prompt.

The label-conditioned analogical-regression baseline implements a stronger deterministic alternative. It fits separate through-origin slopes to City A and City B, uses City C’s coverage label to select the reference assigned to the same response regime, and predicts with that slope. At $k > 0$, it pools the selected reference rows with the visible City C rows before refitting. In the misleading arm, the inverted label selects the opposite reference. This is the minimal successful policy described in the Methods.

Table 13: **Label-conditioned analogical regression beside the six deployments.** MAE is in poll points; ε is slope MAE. LLM columns give the range across deployments in the corresponding prompt arm.

k	Correct-label analogical OLS			Correct-cue LLMs MAE range	Inverted-label analogical OLS			Wrong-cue LLMs MAE range
	MAE	ε	ρ		MAE	ε	ρ	
0	1.842	.261	.656	1.596–2.474	4.561	.640	-.667	4.204–4.361
1	1.640	.230	.702	1.586–3.300	3.722	.523	-.507	3.147–4.462
2	1.495	.214	.737	1.854–3.600	3.174	.447	-.351	2.593–3.935
3	1.430	.204	.756	1.773–3.047	2.797	.391	-.159	2.409–3.518
4	1.337	.193	.773	1.594–2.809	2.483	.348	-.002	2.100–3.043

At $k = 0$, both analogical curves lie within the corresponding deployment range and reproduce the sign reversal in ρ . If a model matches this baseline, Coin City demonstrates competent structured analogy, not evidence beyond that strategy. If a model beats it, the result rules out this particular simple implementation but does not distinguish context-conditioned analogy from latent-dynamics inference.

B.5.3 UNCERTAINTY AND COMPARISON FAMILIES

Primary arm contrasts use percentile intervals from 2,000 bootstrap resamples of the paired episode-level absolute-error difference. Deterministic, model-specific seeds make these intervals reproducible. The $k = 0$ pairwise model table uses 4,000 paired bootstrap resamples with seed 20260812 and the common episode intersection for each model pair. Negative differences favor the first-listed condition or model.

The episodes are the resampling units. Intervals are pointwise descriptions of this fixed seeded synthetic evaluation: they are not simultaneous bands across prefixes, and no multiple-comparison correction is applied. Accordingly, the count of pairwise intervals excluding zero is an exploratory, unadjusted summary.

B.6 COMPLETE RESPONSE-INFERENCING DIAGNOSTICS

The tables below retain a common arm and prefix ordering. Within a deployment and prefix, all three primary arms use the same complete-case episode set; n is repeated so that every exception is visible. Boldface identifies the lowest forecast MAE or slope MAE, and the highest slope correlation, within that deployment and prefix.

B.6.1 PER-DEPLOYMENT PRIMARY-ARM RECOVERY

Table 14: Claude Opus 4.8 forecast and latent-response diagnostics for every prompt arm. Forecast MAE is in poll points; slope MAE and ρ compare the implied response with the true City C slope.

k	Arm	n	Forecast MAE	Slope MAE	$\rho(\hat{g}, g_C)$
0	Target only	250	2.321	.332	.019
	ABC, no C context	250	2.651	.379	-.063
	ABC + C context	250	1.817	.265	.700
1	Target only	250	3.730	.549	.371
	ABC, no C context	250	2.438	.358	.322
	ABC + C context	250	2.147	.316	.584
2	Target only	250	2.668	.403	.383
	ABC, no C context	250	2.041	.296	.492
	ABC + C context	250	2.079	.308	.491
3	Target only	250	2.485	.358	.449
	ABC, no C context	250	2.128	.305	.513
	ABC + C context	250	1.977	.291	.627
4	Target only	250	2.209	.324	.498
	ABC, no C context	250	1.929	.281	.542
	ABC + C context	250	1.760	.257	.658

Table 15: GPT-5.6 forecast and latent-response diagnostics on the same frozen prompts. Metrics use episodes parseable in all three arms at each prefix.

k	Arm	n	Forecast MAE	Slope MAE	$\rho(\hat{g}, g_C)$
0	Target only	250	3.596	.511	.054
	ABC, no C context	250	2.518	.352	-.041
	ABC + C context	250	1.596	.224	.678
1	Target only	249	2.993	.439	.394
	ABC, no C context	249	2.768	.399	.302
	ABC + C context	249	1.586	.225	.669
2	Target only	250	2.560	.387	.459
	ABC, no C context	250	2.050	.299	.534
	ABC + C context	250	1.854	.268	.606
3	Target only	250	2.240	.328	.528
	ABC, no C context	250	1.967	.286	.584
	ABC + C context	250	1.773	.256	.600
4	Target only	250	2.153	.318	.577
	ABC, no C context	250	1.746	.253	.653
	ABC + C context	250	1.594	.228	.664

Table 16: DeepSeek-V4-Pro forecast and latent-response diagnostics on the same frozen prompts. All 250 episodes parsed in every arm and prefix.

k	Arm	n	Forecast MAE	Slope MAE	$\rho(\hat{g}, g_C)$
0	Target only	250	2.447	.353	-.027
	ABC, no C context	250	2.841	.401	-.027
	ABC + C context	250	2.474	.344	.523
1	Target only	250	3.585	.525	.334
	ABC, no C context	250	3.130	.458	.302
	ABC + C context	250	2.456	.347	.578
2	Target only	250	3.380	.501	.319
	ABC, no C context	250	3.423	.502	.365
	ABC + C context	250	2.687	.391	.492
3	Target only	250	2.908	.424	.454
	ABC, no C context	250	3.224	.463	.403
	ABC + C context	250	2.797	.403	.504
4	Target only	250	2.812	.415	.430
	ABC, no C context	250	3.040	.459	.366
	ABC + C context	250	2.809	.409	.470

Table 17: Kimi K3 forecast and latent-response diagnostics on the same frozen prompts. All 250 episodes parsed in every arm and prefix.

k	Arm	n	Forecast MAE	Slope MAE	$\rho(\hat{g}, g_C)$
0	Target only	250	3.352	.476	-.016
	ABC, no C context	250	2.548	.357	-.021
	ABC + C context	250	1.886	.266	.593
1	Target only	250	3.362	.494	.365
	ABC, no C context	250	3.370	.490	.345
	ABC + C context	250	2.382	.341	.579
2	Target only	250	3.194	.491	.387
	ABC, no C context	250	3.114	.487	.352
	ABC + C context	250	2.306	.349	.455
3	Target only	250	2.816	.412	.430
	ABC, no C context	250	2.636	.407	.410
	ABC + C context	250	2.005	.292	.609
4	Target only	250	2.378	.357	.444
	ABC, no C context	250	1.957	.284	.622
	ABC + C context	250	1.840	.263	.601

Table 18: Gemini 3.6 Flash forecast and latent-response diagnostics on the same frozen prompts. All 250 episodes parsed in every arm and prefix.

k	Arm	n	Forecast MAE	Slope MAE	$\rho(\hat{g}, g_C)$
0	Target only	250	2.534	.370	.002
	ABC, no C context	250	2.552	.356	.078
	ABC + C context	250	2.174	.302	.559
1	Target only	250	3.733	.550	.403
	ABC, no C context	250	3.717	.548	.361
	ABC + C context	250	3.300	.500	.292
2	Target only	250	3.564	.558	.348
	ABC, no C context	250	3.703	.579	.335
	ABC + C context	250	3.600	.569	.330
3	Target only	250	3.370	.513	.314
	ABC, no C context	250	3.092	.460	.397
	ABC + C context	250	3.047	.443	.471
4	Target only	250	3.308	.498	.389
	ABC, no C context	250	2.911	.447	.433
	ABC + C context	250	2.793	.419	.447

Table 19: Claude Opus 5 forecast and latent-response diagnostics on the same frozen prompts. Two responses exhausted the output budget on extended thinking and returned no answer, one in each ABC arm.

k	Arm	n	Forecast MAE	Slope MAE	$\rho(\hat{g}, g_C)$
0	Target only	250	2.359	.333	-.018
	ABC, no C context	250	2.619	.371	-.007
	ABC + C context	250	1.764	.247	.617
1	Target only	249	3.549	.520	.391
	ABC, no C context	249	2.303	.338	.370
	ABC + C context	249	1.741	.254	.601
2	Target only	250	3.804	.596	.309
	ABC, no C context	250	2.680	.412	.389
	ABC + C context	250	2.060	.314	.524
3	Target only	250	3.022	.461	.379
	ABC, no C context	250	2.363	.356	.459
	ABC + C context	250	2.075	.309	.550
4	Target only	249	2.700	.418	.443
	ABC, no C context	249	2.369	.366	.459
	ABC + C context	249	2.201	.335	.499

B.6.2 MISLEADING-CUE STRESS TEST

Table 20: Misleading-cue arm: forecast MAE and latent-slope correlation at each City C prefix. The prompt is byte-identical to the correct-context arm except that the City C sentence names the opposite response regime. Negative ρ means the implied response is anti-correlated with the truth, i.e. the deployment has adopted the stated regime. All six deployments completed all 1,250 prompts in this arm.

Model	$k=0$		$k=1$		$k=2$		$k=3$		$k=4$	
	MAE	ρ	MAE	ρ	MAE	ρ	MAE	ρ	MAE	ρ
Claude Opus 4.8	4.20	-.66	3.88	-.51	3.04	.09	2.61	.22	2.31	.34
GPT-5.6	4.23	-.68	4.15	-.63	2.78	.17	2.71	.25	2.10	.46
DeepSeek-V4-Pro	4.36	-.62	4.46	-.48	3.61	.07	3.39	.14	3.04	.32
Kimi K3	4.31	-.62	4.12	-.25	3.03	.15	2.72	.14	2.18	.44
Gemini 3.6 Flash	4.29	-.50	4.40	-.06	3.93	.24	3.52	.25	2.93	.36
Claude Opus 5	4.25	-.62	3.15	-.21	2.59	.27	2.41	.41	2.36	.47

B.6.3 ARBITRARY-SYMBOL INDUCTION CONTROL

Table 21: **Episode-randomized arbitrary-symbol control for DeepSeek-V4-Pro.** All three arms were collected contemporaneously on the same deployment resource. MAE is forecast error in poll points; ρ is the correlation between implied and true latent slopes; Acc. classifies the higher- versus lower-response regime at the midpoint 0.575. Each Δ is symbol minus no context, with a paired 95% episode-bootstrap interval. At $k = 0$, the City C label can be decoded only from City A/B examples.

k	n	No C context		Semantic context		Arbitrary symbol				Symbol – no context		
		MAE	ρ	MAE	ρ	MAE	ρ	Acc.		Δ MAE [CI]	$\Delta\rho$ [CI]	Δ Acc. [CI]
0	250	2.62	0.02	2.59	0.42	2.73	0.45	0.69	0.11	[-0.22, 0.45]	0.44 [0.32, 0.56]	0.17 [0.10, 0.24]
1	250	3.03	0.33	2.29	0.59	2.82	0.47	0.74	-0.21	[-0.55, 0.12]	0.14 [0.04, 0.25]	0.09 [0.03, 0.15]
2	250	3.13	0.36	2.65	0.53	2.71	0.42	0.74	-0.42	[-0.82, -0.05]	0.06 [-0.05, 0.17]	0.05 [-0.00, 0.11]
3	250	3.20	0.38	2.71	0.49	2.74	0.50	0.77	-0.46	[-0.79, -0.16]	0.11 [0.03, 0.20]	0.12 [0.07, 0.19]
4	250	2.81	0.43	2.74	0.50	2.64	0.46	0.71	-0.17	[-0.45, 0.10]	0.03 [-0.05, 0.12]	0.01 [-0.05, 0.06]

At $k = 0$, where City C has no numerical examples, the arbitrary label raises latent-slope correlation from .018 without context to .454 ($\Delta\rho = .436$, 95% CI [.323, .558]) and regime accuracy from .520 to .692 ($\Delta = .172$, 95% CI [.104, .240]). The semantic arm reaches $\rho = .422$ and accuracy .704 in the same rerun. Thus DeepSeek can recover which label denotes which response regime from City A/B examples alone; a fixed semantic prior that national coverage implies larger effects cannot explain this discrimination.

The control does not produce a detectable zero-shot MAE gain: symbol minus no context is +.107 poll points (95% CI [-.222, .452]). This is not contradictory: the no-context arm can attain moderate absolute error by predicting a central slope while failing to distinguish the two regimes. With direct City C rows, the symbol arm lowers MAE at $k = 2$ (-.420, 95% CI [-.817, -.047]) and $k = 3$ (-.458, 95% CI [-.789, -.157]); its ρ remains .418-.498 across all prefixes. These results rule out the specific lexical-prior-only account for this deployment, but numerical analogy plus episode-local code induction remains a sufficient alternative explanation.

B.6.4 PAIRED CUE EFFECTS AND INTERPRETATION

Table 22: Paired forecast-MAE difference between ABC plus City C context and ABC without City C context. Negative values favor context.

Model	k	Mean paired difference	95% bootstrap CI
Claude	0	-.834	$[-1.086, -.597]$
	1	-.292	$[-.544, -.048]$
	2	.038	$[-.155, .254]$
	3	-.151	$[-.352, .035]$
	4	-.169	$[-.343, .001]$
GPT-5.6	0	-.922	$[-1.151, -.686]$
	1	-1.182	$[-1.470, -.925]$
	2	-.197	$[-.331, -.067]$
	3	-.193	$[-.355, -.049]$
	4	-.152	$[-.285, -.029]$
DeepSeek-V4-Pro	0	-.366	$[-.679, -.046]$
	1	-.674	$[-1.016, -.360]$
	2	-.736	$[-1.111, -.376]$
	3	-.427	$[-.744, -.116]$
	4	-.230	$[-.601, .091]$
Kimi K3	0	-.662	$[-.923, -.401]$
	1	-.987	$[-1.308, -.680]$
	2	-.808	$[-1.132, -.470]$
	3	-.631	$[-.910, -.359]$
	4	-.117	$[-.291, .065]$
Gemini 3.6 Flash	0	-.378	$[-.689, -.052]$
	1	-.417	$[-.682, -.139]$
	2	-.103	$[-.337, .108]$
	3	-.045	$[-.392, .303]$
	4	-.118	$[-.378, .162]$
Claude Opus 5	0	-.856	$[-1.087, -.602]$
	1	-.562	$[-.733, -.410]$
	2	-.620	$[-.797, -.461]$
	3	-.288	$[-.443, -.139]$
	4	-.168	$[-.276, -.071]$

The zero-shot matched-reference comparison shows that the target-city description controls how the A/B cases are used. All six $k = 0$ bootstrap intervals exclude zero. The analogical-regression baseline shows that this result is fully consistent with direct label-conditioned reference selection and slope transfer. More target rows do not monotonically reduce error: DeepSeek and Opus 5 drift by 0.34 and 0.44 points from $k = 0$ to $k = 4$, and Gemini’s context advantage is significant only at $k \in \{0, 1\}$. Short, noisy prefixes can temporarily mislead either arm.

B.7 MISSINGNESS, SCOPE, AND ARTIFACT MAP

B.7.1 MISSING RESPONSES AND FILE-LEVEL REPAIRS

Claude Opus 5 has two unparseable primary task cells: one correct-context response at $k = 1$ and one no-context response at $k = 4$. Both exhausted the output budget during extended thinking and returned no answer. GPT-5.6 has one no-context failure at $k = 1$, where the provider returned a model-side HTTP 500 before any response. None was rerun. Complete-case pairing therefore gives $n = 249$ for the corresponding deployment-prefix rows and $n = 250$ elsewhere. These three failures constitute 0.013% of the 22,500 primary task cells.

Claude Opus 4.8, DeepSeek-V4-Pro, Kimi K3, and Gemini 3.6 Flash have 3,750 parseable primary responses each after the one offline Claude key-whitespace normalization described above. That normalization changed neither the forecast value nor the raw model answer and made no additional API call.

All six misleading-context analytic files contain exactly 1,250 unique parseable tasks. Concurrent writes in the Kimi collection produced 11 duplicate records and 14 malformed intermediate lines. Construction of the analytic file retained the earliest response for each task and identified 11 absent task IDs; those 11 tasks were collected under the unchanged Kimi configuration. Thus 7,500 is the number of final unique analytic records for the misleading arm, not the number of intermediate records or provider-level attempts.

The arbitrary-symbol evaluation contains 1,250 symbol, 1,250 contemporaneous no-context, and 1,250 contemporaneous semantic-context DeepSeek records. All 3,750 task IDs are unique, parseable, and matched to their task-file prompt hashes. This is a complete DeepSeek control, not a six-deployment replication.

B.7.2 INTERPRETIVE SCOPE

The experiment establishes context-conditioned response-regime assignment and integration of direct City C evidence in a controlled synthetic setting. It does not identify an explicit latent-dynamics representation. Label-conditioned analogical regression is sufficient: a forecaster can map the coverage label to the matching reference, estimate that reference’s slope, and transfer it to City C. The misleading arm establishes causal dependence on the label, not which internal representation implements that dependence.

The primary binary semantic cue is aligned by construction with the higher- or lower-response family and has a fixed meaning across episodes. The arbitrary-symbol control removes that fixed lexical meaning for one deployment by reversing the KIV/ZOR mapping across episodes; it therefore addresses whether the semantic-arm result can arise solely from a pre-existing national-greater-than-local prior. It does not establish that all six deployments induce new mappings, nor does it establish performance with ambiguous, continuous, or adversarial real-world cues. All prompts also tell models that within-city response is stable and explain the matched-pair structure.

The query target is a latent conditional mean with no newly sampled observation noise. This isolates response recovery but is not equivalent to calibration on a future noisy poll. The six deployments also differ in provider, reasoning mode, output budget, and service tier; model-to-model comparisons are paired but are not configuration-matched capability rankings. The misleading-context arm was added after the primary three-arm freeze and should be interpreted as a stress test, not part of the pre-call primary freeze.

Finally, the 250 episodes are one reproducible seeded synthetic population. Bootstrap intervals resample episodes and are pointwise; they do not incorporate generator-parameter uncertainty, provider drift, or a correction for the many model, arm, and prefix comparisons. Follow-up work should vary cue strength, add a semantic cue-only arm without A/B examples, replicate the arbitrary-symbol control across deployments, evaluate additional generated populations, and score predictive calibration against noisy held-out outcomes.

B.7.3 REPRODUCIBILITY AND ARTIFACT MAP

The frozen data root is `exp2_v2/biased_news/data/coin_city_stable_relationship_claude_n250_v4/`, and the experiment-code root is `exp2_v2/biased_news/`. Paths below are relative to those roots unless a repository-level path is shown.

Design and preflight. Under the data root, `design/manifest.json` and `design/validation.json` freeze and validate the primary design; `design/tasks_wrong_context.manifest.json` records the misleading arm, and `design/tasks_abc_symbol_context.manifest.json` records the balanced, episode-randomized symbol control.

Generator and task builders. Under the code root, the generator is `engine/coin_city_stable_relationship_claude_n250.py`; the builders are `eval/build_coin_city_stable_relationship_claude_n250.py` and `eval/build_coin_city_wrong_context_tasks.py`. The symbol implementation is `engine/coin_city_symbol_context_arm.py`, with task builder `eval/build_coin_city_symbol_context_tasks.py`.

Execution and parsing. The manifest-locked Opus 4.8 wrapper is `eval/run_coin_city_stable_relationship_claude_n250.py`; shared append-only execution and parsing are in `eval/run_frozen_task_shard.py`. Model-specific runners and `eval/run_coin_city_wrong_context.py` preserve configurations and response records under the data root. The symbol runner is `eval/run_coin_city_symbol_context.py`; its contemporaneous comparison arms are under `responses/symbol_control_matched_20260813/`.

Primary analysis. `analysis/analyze_coin_city_stable_relationship_multimodel.py` under the code root produces `analysis/multimodel_results.json` under the data root, the authority for the primary-arm metrics and paired contrasts.

Arbitrary-symbol analysis. `analysis/analyze_coin_city_symbol_context.py` produces `analysis/symbol_context_results.json`; focused design checks are in `tests/test_coin_city_symbol_context.py`.

Misleading arm and paper outputs. The repository-level script `paper/generate_coin_city_split_figures.py` recomputes misleading-arm metrics from the final response files and scoring key. `paper/generate_coin_city_appendix_tables.py` generates the primary tables and pairwise contrasts.

C EXPERIMENTS 3 AND 4: TRAINING AND TRANSFER

This appendix reports the complete Coin City A-to-B design and Qwen3-4B endpoints, the completed 48-endpoint structural-OOD factorial, and the Experiment 4 historical-market protocol and results. It also records the effective training and evaluation configuration for every reported cell.

C.1 EXPERIMENT 3: COIN CITY DOMAIN AND STRUCTURAL TRANSFER

C.1.1 DATA-GENERATING PROCESSES

Training uses only Structure A, the direct relationship from Experiment 2. On the displayed scale of domain d ,

$$y_{t+1} = b_d + g_j x_t + \epsilon_t, \quad \epsilon_t \sim \mathcal{N}(0, \sigma_d^2). \quad (6)$$

A pulse acts in the next period and has no residual effect under zero subsequent input. The unseen evaluation mechanism, Structure B, is

$$m_{t+1} = \rho m_t + g_j x_t, \quad (7)$$

$$y_{t+1} - b_d = \phi(y_t - b_d) + \lambda m_{t+1} + \epsilon_t. \quad (8)$$

This is a linear-Gaussian latent-state process in the sense of [Kalman \(1960\)](#): it introduces a persistent mediator and outcome carry-over, so a single pulse has both immediate and delayed effects. Model prompts disclose neither equation, parameter values, internal variable, nor structure label.

Coin City uses $b = 50$, range $[0, 100]$, driver pool $\{-12, -8, -4, 0, 4, 8, 12\}$, and $\sigma = 1.6$. Held-out Coin Harbor uses $b = 180$, range $[40, 320]$, driver pool $\{-24, -16, -8, 0, 8, 16, 24\}$, and $\sigma = 4.5$, together with new terminology, units, and qualitative descriptions. Evaluation gains follow $g \sim \text{Uniform}(.22, .92)$. Structure B independently samples $\rho \sim \text{Uniform}(.48, .74)$, $\phi \sim \text{Uniform}(.18, .42)$, and $\lambda \sim \text{Uniform}(.68, .94)$. A reference and its matched target share these parameters but have independent inputs and noise.

Table 23: **Exact Coin City–Coin Harbor domain contrast.** Both surfaces instantiate the same Structure A and B simulators and task template; Coin Harbor changes the semantic setting, time unit, numeric scale, interventions, noise, and pathway descriptions.

Field	Coin City (train and test)	Coin Harbor (test only)
Driver $\rightarrow$ outcome	Net campaign news $\rightarrow$ candidate support	Net shipping bulletin $\rightarrow$ cargo-flow index
Time unit	Week	Tide cycle
Resting level / support	50 / [0, 100]	180 / [40, 320]
Observed-input pool	$\{-12, -8, -4, 0, 4, 8, 12\}$	$\{-24, -16, -8, 0, 8, 16, 24\}$
Evaluation shocks	$\{-12, -6, 0, 6, 12\}$	$\{-24, -12, 0, 12, 24\}$
Observation-noise SD	1.6 support points	4.5 cargo-index points
Structure A description	Synchronized citywide alerts; reactions occur mostly in the same week	Synchronized fleetwide radio; responses occur mostly in the same tide cycle
Structure B description	Neighborhood conversations and local organizations; reactions build gradually and persist	Dockside crews and harbor groups; responses build gradually and persist

C.1.2 TRAINING EPISODES AND MATCHED CONTROL

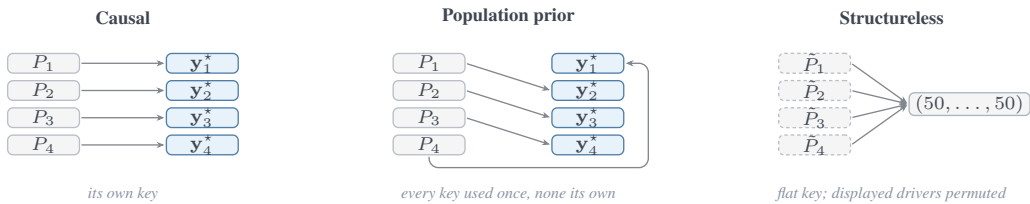
Each training prompt contains two 14-case Coin City reference systems, both governed by Equation 6. One draws $g \in [.18, .46]$ and the other $g \in [.68, .98]$; their order is randomized. The target shares exactly one reference gain, selected in balanced fashion, and shows $k \in \{0, 2, 4, 8\}$ independently generated cases. Its national/local background description identifies the matching weak or strong response as in Experiment 2. There are 1,200 rows at each k , or 4,800 per training run.

The causal and population-prior datasets have byte-identical prompts. Causal reward uses the target’s noise-free forecasts. For the control, targets are deranged within each k stratum with a fixed permutation: no row keeps its own target, while the complete multiset of target vectors is exactly preserved. Consequently the control can learn the output format and population distribution but cannot improve reward by using that row’s references or target history. The Qwen3-4B structureless diagnostic additionally scrambles displayed drivers and rewards a flat intervention response; it is not a confirmatory comparator.

Table 24: **One training episode, and what each policy is rewarded against.** Panel A is the prompt for episode `train:1:k2`, abridged from the rendered text. Panel B is the prompt–key pairing used during training, drawn for four episodes at one evidence depth: P_i is episode i ’s prompt and y_i^* its ten-answer key. Causal and population-prior training use byte-identical prompts and the same multiset of keys; only the pairing differs. The control shifts cyclically across all 1,200 episodes at a depth, so its key is a real key drawn from the same distribution but never this prompt’s, and no reward can be earned by reading this row’s references. Base is untrained and receives no supervision.

A. Shared prompt. Coin City; resting level 50.0, bounded to [0, 100]; no equation or coefficient is shown.
REFERENCE 1 "...immediate national alerts ...react **strongly**" 14 wk: $-8.0 \rightarrow 43.8, -12.0 \rightarrow 40.9, \dots, 0.0 \rightarrow 51.6$
REFERENCE 2 "...local civic briefings ...react **weakly**" 14 wk: $+8.0 \rightarrow 52.6, +4.0 \rightarrow 51.3, \dots, -12.0 \rightarrow 46.9$
TARGET ($k=2$) "...immediate national alerts ...react **strongly**" $-8.0 \rightarrow 42.3, +4.0 \rightarrow 53.9$
QUERY ten counterfactuals A–J: shock $\in \{-12, -6, 0, 6, 12\} \times \text{horizon} \in \{1, 3\}$; return {"forecasts": [A, ..., J]}

B. How prompts are paired with answer keys during training.



C.1.3 EVALUATION INTERFACE AND QUALITATIVE HINT

Every evaluation prompt contains a complete 14-period Structure A reference, a complete 14-period Structure B reference, and a target governed by one of them. For Coin City the two reference descriptions are:

Residents receive campaign developments through synchronized citywide alerts. Reactions occur mostly during the same week as the news.

Campaign developments usually spread through neighborhood conversations and local organizations. Reactions may build gradually and persist after the original news has faded.

Coin Harbor uses the matched fleetwide-radio versus dockside-network descriptions. The target receives the correct description, no description, or the other pathway’s description. The three prompt variants reuse the exact reference and target trajectories, scenario grid, and gold forecasts; the freeze records a common numeric hash for every cue triplet.

Each task requests ten counterfactuals: shocks $\{-2u_d, -u_d, 0, u_d, 2u_d\}$ at horizons one and three, with $u_d = 6$ in Coin City and $u_d = 12$ in Coin Harbor. The shock occurs in the next period and all later inputs are zero. The response vector consists of the eight nonzero-shock minus zero-shock contrasts. The full endpoint contains

$$2 \text{ domains} \times 2 \text{ structures} \times 3 \text{ hints} \times 4 \text{ } k \text{ values} \times 30 \text{ episodes} = 1,440$$

paired prompts. Training and evaluation seed bands begin at 81,000,000 and 91,000,000, respectively.

Exact task template and prompt variants. The following is the task string emitted by the prompt generator before the model-specific chat wrapper is applied. Bracketed uppercase fields below denote row-specific substitutions; they are not shown to the model. Line wrapping is typographic, while the blank-line-separated sections and all fixed wording are part of the prompt.

```
Forecasting task in [DOMAIN].

The resting level of [OUTCOME] is [BASELINE]. Values are bounded to [LOWER,
UPPER].

The reference systems are complete examples. The target system may share a
stable response pattern with one reference. Infer from the information
shown; no response equation or coefficient is provided.

REFERENCE SYSTEM 1
Background: [REFERENCE 1 CONTEXT]
[REFERENCE 1 TABLE: period, driver, and 14 observed outcomes]

REFERENCE SYSTEM 2
Background: [REFERENCE 2 CONTEXT]
[REFERENCE 2 TABLE: period, driver, and 14 observed outcomes]

TARGET SYSTEM
[TARGET BACKGROUND, WHEN PRESENT]
[TARGET TABLE WITH k OBSERVATIONS, OR THE EXACT NO-OBSERVATION SENTENCE]

COUNTERFACTUAL FORECASTS
For each scenario, apply the stated [DRIVER] in the next [PERIOD] and then
set it to 0 in later periods. Forecast [OUTCOME] at the stated horizon.

A: shock=[VALUE]; horizon=1 ... E: shock=[VALUE]; horizon=1
F: shock=[VALUE]; horizon=3 ... J: shock=[VALUE]; horizon=3

Return only strict JSON with exactly ten finite numbers in scenario order:
{"forecasts": [A,B,C,D,E,F,G,H,I,J]}
```

The renderer realizes the variants as follows. Training always includes the correct strong/weak background and uses the same input string for causal and population-prior training; only the hidden reward target differs. At evaluation, correct inserts the target pathway’s description, none omits the background line entirely, and misleading inserts the other pathway’s description. Every cue triplet otherwise has identical tables, interventions, and targets. If $k = 0$, the target block contains the exact sentence *No target-system observations are available yet.*; otherwise it contains the same table format as the references with the first k rows. The Qwen3-4B structureless

diagnostic retains this shell but swaps every displayed $-12.0 \leftrightarrow 12.0$ and $-8.0 \leftrightarrow 8.0$ token through a collision-safe substitution. Untrained bases receive the same held-out prompts as trained models. Thus cue, reward-arm, structureless, and base comparisons do not introduce an unreported instruction change.

C.1.4 ESTIMANDS, ROSTER, AND VALIDATION CRITERIA

The confirmatory grid is three models (Qwen3-4B, Qwen3-8B, and Llama-3.1-8B), two arms, and three training seeds, for 18 runs. Three Qwen3-4B structureless runs and three untrained endpoints are diagnostic. The primary estimand is population-prior minus causal response MAE on Structure B in Coin Harbor with the correct hint, averaging the four k values equally. Positive values favor causal training. Prespecified secondary contrasts apply the same estimator to Structure A/Coin Harbor (domain transfer), Structure B/Coin City (structural transfer), and the in-distribution cell. Paired absent-minus-correct and misleading-minus-correct contrasts measure hint use. Their paired $k = 8$ minus $k = 0$ difference tests whether observations override a misleading prior; negative values indicate a shrinking misleading-hint penalty.

Greedy output is retained for continuity, and every checkpoint adds five draws at temperature .7. Draws are averaged within episode before inference. The hierarchical bootstrap resamples the three training seeds at the top level and paired episodes within each selected seed. Parse failures receive the full observable-range error penalty rather than being discarded.

Before training, the dataset passed 17 fail-closed checks: exact row counts and factorial balance; byte-identical causal/prior prompts and held-out datasets; exact within- k target marginals with no fixed reward links; absence of Structure B from training prompts and metadata; cue-triplet numeric identity; train/evaluation identifier disjointness; hidden-equation leak scans; strict gold round trips; and oracle separation from the blind pathway mixture. Mean A–B response separation is 2.65 points in Coin City and 4.80 in Coin Harbor. Across all three tokenizers, the longest prompt is 1,071 tokens against the registered 2,304-token limit.

The analysis plan assigns population-prior-minus-causal intervals to all four transfer cells, paired intervals to both cue penalties at every k , and the paired $k = 8$ minus $k = 0$ change in misleading-minus-correct error to evidence override. The protocol archive records versioned input, analysis, and output hashes. Full optimization and compute parameters are in Section C.3.

C.1.5 ROUND-300 QWEN3-4B RESULTS

All nine Qwen3-4B causal, population-prior, and structureless runs have produced the complete 1,440-task greedy evaluation and the registered five-draw stochastic evaluation at round 300; the untrained Qwen3-4B endpoint is also complete under both regimes. The task universe is identical across conditions, and parse rates are 100% for every trained arm and 99.97% for the stochastic base. The analysis averages the four evidence depths equally and, for stochastic decoding, averages draws within task before the seed-first bootstrap. The two larger-model Coin City grids were incomplete when this analysis was finalized.

Under five-draw stochastic decoding, the joint-cell MAEs are 6.15, 6.76, 4.81, and 4.49 for base, structureless, population-prior, and causal training, respectively. The prior-minus-causal difference is .319 [.134, .489]; causal training has the lowest point estimate in all four cells, with all four intervals excluding zero. The stochastic nearest-pathway Structure B accuracy is zero for causal training and .873 for base. Table 25 reports the complete transfer grid.

Table 25: **Qwen3-4B Coin City round-300 stochastic endpoint.** Entries are response MAE under the correct hint after averaging five draws within task. Intervals use the registered paired seed-first bootstrap; lower is better.

Test cell	Base	Structureless	Prior	Causal	Prior — causal [95% interval]
City / A (ID)	3.82	2.56	0.79	0.58	0.21 [0.11, 0.32]
Harbor / A (domain)	8.53	5.23	1.91	1.68	0.23 [0.04, 0.38]
City / B (structure)	2.66	3.59	2.53	2.34	0.19 [0.02, 0.35]
Harbor / B (joint)	6.15	6.76	4.81	4.49	0.32 [0.13, 0.49]

The qualitative hint also affects the held-out joint cell. Averaged over evidence depths, omitting the hint raises causal response MAE by .151 [.079, .235], and a misleading hint raises it by .313 [.228, .415]. The misleading-hint penalty is smaller at $k = 8$ than at $k = 0$, but the evidence-override contrast $-.116 [-.309, .100]$ includes zero. Table 26 reports every prespecified hint-by-depth estimate.

Table 26: **Qwen3-4B hint use in the Coin Harbor/Structure B joint-transfer cell under five-draw stochastic decoding.** MAE averages draws within task and training seeds equally. Positive contrasts mean that removing or reversing the qualitative hint increases error. The evidence-override row is the $k = 8$ minus $k = 0$ change in the misleading-minus-correct penalty.

k	Correct	Absent	Misleading	Absent — correct [95% CI]	Misleading — correct [95% CI]
0	4.491	4.672	4.890	.181 [.012, .370]	.399 [.204, .607]
2	4.444	4.530	4.688	.086 [-.064, .271]	.243 [.115, .378]
4	4.527	4.684	4.855	.157 [.037, .289]	.327 [.165, .511]
8	4.510	4.692	4.793	.182 [.033, .331]	.283 [.146, .464]
All k	4.493	4.645	4.806	.151 [.079, .235]	.313 [.228, .415]
Evidence override: change in misleading penalty, $k = 8$ minus $k = 0$					-.116 [-.309, .100]

Greedy decoding yields joint-cell MAEs of 6.09, 6.69, 4.77, and 4.59, respectively, and a prior-minus-causal difference of .174 [.039, .316]. Causal training has the lowest point estimate in all four cells. The interval excludes zero in the in-distribution and joint cells, but includes zero in the domain-only and structure-only cells, marginally so for structure-only ($[-.002, .248]$). Correct, absent, and misleading hints yield causal MAE of 4.59, 4.68, and 4.86 in the joint cell. Its nearest-pathway Structure B accuracy is .025, versus .817 for base.

Table 27: **Qwen3-4B Coin City round-300 greedy endpoint.** Entries are response MAE under the correct hint. Intervals use the same registered paired seed-first bootstrap; lower is better.

Test cell	Base	Structureless	Prior	Causal	Prior — causal [95% interval]
City / A (ID)	3.76	2.56	0.66	0.55	0.11 [0.07, 0.15]
Harbor / A (domain)	8.16	5.23	2.30	2.12	0.18 [-0.14, 0.45]
City / B (structure)	2.55	3.59	2.37	2.23	0.14 [-0.002, 0.248]
Harbor / B (joint)	6.09	6.69	4.77	4.59	0.17 [0.04, 0.32]

Across both decoding regimes, lower response error does not imply recovery of the mediated shape. The Coin City cross-model endpoint is excluded because its Qwen3-8B and Llama-3.1-8B causal/prior grids were incomplete when this analysis was finalized.

C.2 EXPERIMENT 4: HISTORICAL POLYMARKET PASTCASTING

This study asks whether outcome-based training transfers to events and normalized question families unseen during training, using a point-in-time cutoff and sealed test.

C.2.1 FROZEN SAMPLE AND LEAKAGE GATES

Frozen source and sample. The source export² has 1,788,703 rows and SHA-256 5697f767...f638e7d (the full digest is stored in the manifest). The builder scans the entire file and fails on checksum drift. The frozen calendar splits contain 1,736 train, 512 development, and 1,024 test markets, with YES rates .311, .275, and .301. Train contains 991 distinct events and 1,456 normalized question templates; development contains 292 and 453; test contains 493 and 866. The selected archive rows have no official series identifiers, so event and template components supply the operative family groups.

Eligibility and point-in-time censoring. We admit literal Yes/No markets resolved by an extreme final token price. A boundary-aware domain rule selects political, governmental, macroeconomic, and geopolitical questions while explicitly rejecting culture, awards, sports, esports, and crypto categories. This rule is fixed before model training. One task is built 30 days before the earliest archived terminal timestamp. At least four distinct UTC dates must have genuine pre-cutoff YES candles, the latest candle must be no more than seven days stale, and its price must remain in [.02, .98]. We keep at most the latest 14 dates. Empty, carry-forward, and post-cutoff candles are rejected. The prompt omits the outcome, final prices and volume, market status, realized terminal time, and the tags used for domain selection.

The archive records timestamped price history but not revisions to static question or resolution-rules text. We therefore treat those strings as time-invariant. This is a residual point-in-time limitation: the audit can prove that no timestamped or outcome field crosses the cutoff, but cannot prove that an archived rule was never edited later.

Family-disjoint temporal split. Before applying sample caps, every development row sharing an event, available series, or normalized template with train is removed; the same rule removes test rows matching train or development. Numeric values and month names are normalized in template keys. The final intersection is zero for each pair of splits. Sampling within a split is a stable hash of task identifier and protocol version, so row order in the export cannot change membership.

C.2.2 REGISTERED TRAINING AND SEALED EVALUATION

Matched update budget. The target domain yields 1,736 unique train markets rather than 4,800. We preserve the precommitted 300-round comparison with Experiment 3A using a deterministic, balanced schedule of 4,800 presentations: every market appears twice and 1,328 appear a third time. Schedule order and task files are hash-locked. With batch size 16 this gives exactly 300 rollout-and-update rounds (2,400 AdamW steps). This distinction between unique markets and presentations is retained in every manifest and table. The longest wrapped prompt is 1,012 Qwen tokens against the 1,792-token training limit, so OAT’s prompt filter drops no task.

Table 28: Experiment 4 training configuration. Test data are absent from the training process and evaluated once after all three training seeds succeed.

Quantity	Value
Base / adapter	Qwen3-4B-Instruct-2507; LoRA rank 32, $\alpha = 64$
Algorithm	Dr. GRPO; $\beta = 0$; AdamW, lr 10^{-6}
Unique train / presentations	1,736 / 4,800 balanced repetitions
Rounds / seeds	300; 42, 43, 44
Rollout	16 prompts, 8 samples each; temperature 1.3
Output	one JSON probability; 128 new-token limit
Reward	$1 - (\hat{p} - y)^2$ only
Development	512 tasks; greedy evaluation every 25 rounds
Locked test	1,024 tasks; greedy, once after all seeds succeed

C.2.3 COMPLETED LOCKED-TEST ACCURACY

Frozen comparators and decision rule. The latest genuine market price has test Brier .11369 and log loss .36581. Platt calibration fit on train only uses intercept .03166 and market-logit slope

² 🤖 Full Polymarket archive. Link anonymized for review.

1.20747; it obtains test Brier .11232 and log loss .36046. Its Brier improvement over the raw market is $-.00137$ with family-bootstrap 95% interval $[-.00262, -.00011]$. Train prevalence obtains .21042 Brier. These deterministic comparators were frozen before learned-model results. The primary learned contrast is the mean loss over three trained seeds minus base loss. Invalid JSON receives Brier loss one. We report trained minus raw market and trained minus train-only Platt separately, using 5,000 bootstrap resamples of connected event/template components.

All test forecasts parse. The three-seed trained mean obtains Brier .11589 versus .13160 for the base, a difference of $-.01572$ with 95% interval $[-.02528, -.00667]$. Its difference from the raw market is .00219 with interval $[-.00264, .00721]$, and its difference from train-only Platt is .00356 with interval $[-.00125, .00874]$. Thus the learned model passes the primary base-relative decision but not the stronger market-improvement criterion. The result establishes transfer relative to the untrained model, not an advantage over the market signal.

Table 29: **Per-seed locked-test results.** Every trained seed improves on the untrained Brier score of .1316; the main text reports only the predeclared three-seed mean.

Forecaster	Updates	Brier ↓	Δ base	Δ market	Log loss ↓
Trained, seed 42	300	.1142	-.0174	+.0005	.3780
Trained, seed 43	300	.1140	-.0176	+.0003	.3772
Trained, seed 44	300	.1194	-.0122	+.0058	.4330
Trained mean	300	.1159	-.0157	+.0022	.3960

C.2.4 SECONDARY EVIDENCE-UPDATING ANALYSIS

After the locked-test decision, we apply the exact Experiment 1 update prompt and all nine frozen packets for each of its 100 markets (three pro- H_1 , three anti- H_1 , and three orthogonal). The packet and initial-context SHA-256 digests are 6741be0a...8796d36 and c550cf05...0c098. Every condition receives the same canonical first structured Qwen 2.5-7B forecast for each market, so this paired analysis isolates conditional revision rather than web retrieval or initial event-model construction. Search and thinking are disabled; decoding is greedy with a 1,500-token output cap. The longest prompt is 1,480 tokens against a 6,144-token context limit.

The paired evaluation produces all 900 packet-conditioned revisions for the base and each final adapter, with 100% parse coverage and ICS equal to one. Base EHC/HFC is .909 on 591 eligible directional updates. Seeds 42, 43, and 44 obtain .917, .922, and .918 (592, 588, and 591 eligible updates), for a three-seed mean of .919. The paired market-bootstrap trained-minus-base difference is $+.0101$ with 95% interval $[+.0004, +.0207]$ for both metrics. Sensitivity is $4.64\times$ for base and averages $4.83\times$ after training; its difference of $+.19$ has interval $[-.07, +.48]$. The secondary analysis therefore detects a small improvement in directional consistency, but not selectivity. It does not alter the locked-test decision and is a secondary updating result, not a second transfer result.

Table 30: **Per-seed evidence-updating results.** The main text reports only the three-seed mean. All conditions use identical fixed contexts and evidence packets.

Condition	Updates	EHC ↑	HFC ↑	Sens. ↑	Valid
Trained, seed 42	300	.917	.917	$4.69\times$	900/900
Trained, seed 43	300	.922	.922	$5.13\times$	900/900
Trained, seed 44	300	.918	.918	$4.68\times$	900/900
Trained mean	300	.919	.919	$4.83\times$	900/900

C.3 TRAINING AND EVALUATION CONFIGURATION

Tables 31–39 record the effective OAT/Dr. GRPO arguments for the reported Coin City, structural-OD, and Polymarket studies, including framework defaults omitted from the command line. Execution manifests hash the data, trainer, launcher, evaluator, prompt renderer, and each cell’s environment.

Step accounting. A “step” in the figures and earlier tables is one rollout-and-update round, not one AdamW call. Each full run consumes 16 prompts per round and samples eight responses per prompt (128 trajectories). With one learner GPU, per-device microbatch one, one PPO epoch, and gradient accumulation 16, OAT performs eight AdamW parameter updates per round. Thus 4,800 prompt presentations give 300 reported rounds and 2,400 AdamW updates. The command line supplied a .03 warmup ratio, but OAT selected Transformers’ `constant` scheduler, which does not use the warmup argument. The effective learning rate was therefore 10^{-6} from the first optimizer update. We use “round” below to avoid counting the eight within-round parameter steps as independent data exposures.

Table 31: **Shared Dr. GRPO implementation parameters.** These values apply to Coin City, Polymarket, and the expanded structural-OD study; the study-specific tables below record exceptions.

Quantity	Effective value
Trainable parameters	Frozen base model; LoRA rank 32, $\alpha = 64$, dropout 0, no bias; targets <code>q.proj</code> , <code>k.proj</code> , <code>v.proj</code> , <code>o.proj</code> , <code>down.proj</code> , <code>up.proj</code> , and <code>gate.proj</code> ; no quantization.
Algorithm	OAT Dr. GRPO (<code>critic_type=drgrpo</code>); eight-response group advantage $r_i - \bar{r}$ with no group-standard-deviation division; policy loss is token-summed with the generation limit as a fixed normalizer (no response-length normalization).
Policy objective	PPO ratio clipping 0.2; truncated importance-sampling cap 2.0; one PPO epoch; KL coefficient $\beta = 0$; no critic/value model; reward whitening, advantage whitening, and REINFORCE mode all off.
Optimizer	PyTorch AdamW (Loshchilov & Hutter, 2019) (substituted for DeepSpeed fused/CPU Adam); learning rate 10^{-6} ; $\beta_1 = .9$, $\beta_2 = .95$; weight decay 0; maximum gradient norm 1.0.
Schedule	Constant 10^{-6} learning rate from the first optimizer update; the supplied .03 warmup ratio is ignored by Transformers’ <code>constant</code> scheduler; one prompt epoch; 4,800 prompt presentations = 300 rollout-and-update rounds = 2,400 AdamW steps.
Batching	16 prompts per round; 8 sampled responses per prompt; 128 trajectories; training microbatch 1; gradient accumulation 16; one learner GPU.
Training sampling	Temperature 1.3; <code>top-p = 1</code> ; <code>top-k = -1</code> ; one independently sampled completion at a time, repeated eight times per prompt.
Truncation and stopping	Model EOS only (no external stop string); a completion ending at the length limit receives reward zero. Missing or invalid required JSON also receives reward zero.
Precision and memory	<code>bfloat16</code> ; FlashAttention 2; gradient checkpointing; DeepSpeed ZeRO-2 (Rajbhandari et al., 2020); reference-model offload; vLLM prefix caching (Kwon et al., 2023); dropout disabled.
Seeds and shuffling	Training seeds 42, 43, and 44. Each dataset is deterministically shuffled once, then OAT shuffles prompt batches independently under the training seed.
Checkpointing and logging	Evaluation includes step zero and the final round; intermediate evaluation at the setting-specific frequency; one final LoRA adapter retained; metrics logged each round; Weights & Biases disabled.

C.3.1 COIN CITY A-TO-B FACTORIAL

Table 32 records the locked configuration for Experiment 3. The Qwen3-4B analysis uses the complete greedy and five-draw stochastic round-300 evaluations from each causal, population-prior, and structureless seed. The full cross-model endpoint additionally requires the Qwen3-8B and Llama-3.1-8B causal/prior grids. The table supplements the shared optimizer details in Table 31; where the two tables differ, this setting-specific table governs. The execution manifest stores the environment; the protocol archive hashes the preflight manifest, data files, generator, prompt renderer, scorer, and trainer entry point.

Table 32: Complete registered configuration for Coin City hidden-structure transfer.

Quantity	Effective registered value
Model roster	Qwen/Qwen3-4B-Instruct-2507, Qwen/Qwen3-8B, and meta-llama/llama-3.1-8B-Instruct.
Chat formatting	Qwen3-4B uses the explicit ChatML wrapper; Qwen3-8B uses its tokenizer template with thinking disabled; Llama uses its tokenizer template. All include an assistant-generation prefix.
Registered grid	3 models $\times$ 2 reward arms (causal, population.prior) $\times$ 3 seeds (42, 43, 44) = 18 full training runs.
Diagnostics	Qwen3-4B structureless at the same three seeds, plus one untrained endpoint for each model. These six diagnostics are excluded from the primary causal-versus-prior estimand.
Training data	Coin City and Structure A only; 4,800 prompts, balanced across $k \in \{0, 2, 4, 8\}$ and weak/strong target match. Two reference trajectories of length 14 and one target trajectory of maximum length 8.
Comparator target construction	Causal reward uses the episode truth. Population-prior reward targets are a fixed no-self-match derangement within each k ; its target-vector multiset is exactly identical to causal training. Prompt strings are byte-identical across these two arms.
Held-out data	2 domains $\times$ 2 target structures $\times$ 3 hint arms $\times$ 4 evidence depths $\times$ 30 episode seeds = 1,440 rows. The complete held-out dataset is byte-identical across all training arms.
Mechanism ranges	Evaluation $g \in [.22, .92]$; Structure B $\rho \in [.48, .74]$, $\phi \in [.18, .42]$, and $\lambda \in [.68, .94]$. Training reference gains are $[\cdot18, \cdot46]$ and $[\cdot68, \cdot98]$.
Domain ranges	Coin City: baseline 50, support $[0, 100]$, noise 1.6, driver pool $\{-12, -8, -4, 0, 4, 8, 12\}$. Coin Harbor: baseline 180, support $[40, 320]$, noise 4.5, driver pool $\{-24, -16, -8, 0, 8, 16, 24\}$.
Interventions	Five shocks at horizons one and three; Coin City $\{-12, -6, 0, 6, 12\}$, Coin Harbor $\{-24, -12, 0, 12, 24\}$. Ten joint forecasts produce eight nonzero-minus-zero response contrasts.
Reward	$\cdot4 r_{\text{level}} + \cdot6 r_{\text{response}}$. Each component averages $\max(0, 1 - e /S)$; level tolerance is 10% of the domain’s observable range (10 Coin City points; 28 Coin Harbor units), and response tolerance is half one shock unit (3 and 6, respectively). Invalid or length-truncated output receives zero online reward.
Output contract	Exactly one JSON object with sole key <code>forecasts</code> and exactly ten finite JSON numbers. The same strict parser is used online and offline; no prose, extra/duplicate key, Boolean, non-finite value, or wrong-length list is accepted.
Structured decoding	Syntax-only XGrammar GBNF in training, online evaluation, endpoint evaluation, and base evaluation. It fixes object shape and arity only; numeric values and their relationship to targets remain unconstrained.
Optimization	Frozen base plus LoRA rank 32, $\alpha = 64$; OAT Dr. GRPO; PyTorch AdamW learning rate 10^{-6} with an effective constant schedule and no warmup; $\beta = 0$; one PPO epoch; gradient norm 1; ZeRO-2 and reference offload.
Batch and update accounting	16 prompts per round; eight rollouts per prompt; 128 trajectories; microbatch 1; gradient accumulation 16; 4,800 presentations = 300 rollout-and-update rounds = 2,400 AdamW steps.
Training generation	Temperature 1.3, top- $p = 1$, top- $k = -1$, model EOS only; 192 new-token limit; prompt limit 2,304; total model context 3,072; vLLM prefix caching enabled.
Precision / memory	BF16, FlashAttention 2, gradient checkpointing, dropout off, no quantization; vLLM GPU fraction .38; expandable-segments CUDA allocator.
Online evaluation	All 1,440 rows at temperature 0, one draw, top- $p = 1$, 192-token limit, batch 120, at step zero and every 50 rounds including round 300.
Endpoint evaluation	Greedy plus five draws per prompt at temperature .7 and top- $p = .95$; trained stochastic decode seed is training seed +20,260,818. Base seeds are 20,260,818 (greedy) and 20,260,819 (stochastic).
Offline scoring	Level MAE, response MAE, response error normalized by the blind 50/50 pathway mixture, response correlation, strict parse coverage, and nearest behavioral pathway. A parse failure receives the full domain range as both level and response error.
Variance and estimands	Five-draw outputs first average within episode. Hierarchical intervals resample three training seeds, then paired episodes. Primary: prior-minus-causal response MAE for correct-hint Structure B in Coin Harbor, equal over k . Domain, structure, joint, hint, and hint-by- k contrasts are prespecified secondary analyses.
Preflight validation	A 10-round run for every model and confirmatory arm must save an adapter, produce the full greedy and stochastic evaluation sets, achieve at least .95 strict parse rate under each decoder, and exhibit nonconstant finite reward. The 17-check data preflight must also pass before estimating the factorial.
Compute	Training uses two NVIDIA A6000 48-GB GPUs (one actor, one learner), 8 CPU cores, and 100 GB host memory. Base and endpoint evaluation uses one A6000, 8 CPU cores, and 24 GB host memory. All runs load models from the same frozen local cache.

C.3.2 STRUCTURAL TRANSFER TO UNSEEN MECHANISM COMPOSITIONS

This study tests broader structural transfer than the two-mechanism Coin City design. Eight balanced training worlds expose individual topology and response primitives. The four held-out blocks contain unseen topology composition, nonlinear composition, mixed threshold–persistence composition, and feedback–saturation with gain and noise beyond the training hull. Disclosed and undisclosed prompts use identical numeric episodes; only the mechanism-description block is present or absent. The completed analysis covers the full Qwen3-4B, Qwen3-8B, and Llama-3.1-8B arm-by-disclosure factorial, the Qwen3-4B structureless diagnostic, and all six untrained endpoints.

Exact disclosed and undisclosed task templates. The prompt generator emits the following raw task string before the model-specific chat wrapper. Bracketed uppercase fields are row substitutions, not text shown to the model. Each calibration and target table is a Markdown table headed by *Step*, the world-specific driver name or names, and the outcome name; inputs are signed to one decimal place and outcomes are printed to two decimal places.

```
You are forecasting [OUTCOME] in [DOMAIN].
Time advances one [PERIOD] at a time. Signed drivers may be positive,
negative, or zero.
The usual outcome level is near [BASELINE]; valid forecasts range from
[LOWER] to [UPPER].
[DISCLOSURE BLOCK]

Calibration trajectory 1 (same response system; its own fixed hidden
sensitivity):
[TEN-ROW CALIBRATION TABLE 1]

Calibration trajectory 2 (same response system; its own fixed hidden
sensitivity):
[TEN-ROW CALIBRATION TABLE 2]

Calibration trajectory 3 (same response system; its own fixed hidden
sensitivity):
[TEN-ROW CALIBRATION TABLE 3]

Target trajectory (infer its own fixed hidden sensitivity):
[TARGET TABLE WITH  $k \in \{3, 6, 9\}$  ROWS]

Starting immediately after the target trajectory, consider each intervention
independently.
The named shock is applied to the first driver for one period; every driver
is zero afterward.
- Scenario A: shock -12; forecast horizon 1 [PERIOD] (s).
- Scenario B: shock -4; forecast horizon 1 [PERIOD] (s).
- Scenario C: shock +0; forecast horizon 1 [PERIOD] (s).
- Scenario D: shock +4; forecast horizon 1 [PERIOD] (s).
- Scenario E: shock +12; forecast horizon 1 [PERIOD] (s).
- Scenario F: shock -12; forecast horizon 3 [PERIOD] (s).
- Scenario G: shock -4; forecast horizon 3 [PERIOD] (s).
- Scenario H: shock +0; forecast horizon 3 [PERIOD] (s).
- Scenario I: shock +4; forecast horizon 3 [PERIOD] (s).
- Scenario J: shock +12; forecast horizon 3 [PERIOD] (s).

Forecast every scenario using one coherent interpretation of the target
trajectory.
Put the ten forecasts in scenario order A through J.
Use exactly the key shown, exactly ten numeric entries, and no prose or extra
keys.
Start your response with {"forecasts":[
Write exactly ten finite decimal numbers separated by commas inside that
list.
After the tenth number, write ]} and stop.
```

For undisclosed, the disclosure block is exactly: *The repeatable response pattern is not described. Infer it from the calibration trajectories and the target trajectory.* For disclosed, the block begins with `[[MECHANISM_DESCRIPTION_BEGIN]]`, ends with `[[MECHANISM_DESCRIPTION_END]]`, and contains a deterministic description assembled from the specified world. It opens with *The stable mechanism is known.* The law sentence is one of: (i) *The immediate driver effect is g times the signed driver value.*; (ii) *The driver effect is $g \cdot s \cdot \tanh(\text{driver}/s)$, so large shocks saturate.*; or (iii) *Driver magnitudes up to t have no effect; beyond that threshold the excess is multiplied by g , with negative effects multiplied by a .* Here s , t , and a are replaced by the values in Table 33.

The next sentence states the path: direct replacement; retained outcome plus current effect; a retained mediator feeding a retained outcome; a feedback state interacting with the driver and outcome; or the composed mediator–feedback path. It prints every retention and coupling value in Table 33. A two-driver world adds exactly *The second driver has a fixed coefficient [VALUE]*. Every disclosed block then ends: *The episode-specific g is fixed within a trajectory and lies between [LOW] and [HIGH]. Observations have Gaussian noise with standard deviation [SD].*

Causal-family and population-prior training receive byte-identical rendered prompts; only their private reward vectors differ. The structureless diagnostic uses the same renderer but independently resamples every displayed calibration and target driver while retaining the displayed outcomes and task identity. Untrained bases receive the same held-out prompt rows as trained models. Thus disclosure is the only instruction-level difference in the confirmatory disclosed/undisclosed comparison.

Table 33: **Frozen mechanism split.** A mechanism identity is the tuple (response law, path topology, number of drivers); domain nouns, baselines, and other surface labels are excluded. No held-out identity occurs in training.

Split / world	Law / path	Dynamic parameters	Gain / noise
Train: direct_linear	linear / direct	—	$g \in [.20, .90];$ $\sigma = 1.2$
Train: persistent_linear	linear / persistent	$\phi_y = .62$	$g \in [.20, .90];$ $\sigma = 1.6$
Train: mediated_linear	linear / mediated	$\phi_y = .22, \phi_m = .48, c = .78$	$g \in [.20, .90];$ $\sigma = 1.4$
Train: feedback_linear	linear / feedback	$\phi_y = .38, \phi_m = .34, f = .36$	$g \in [.20, .90];$ $\sigma = 1.8$
Train: two-driver_linear	linear / persistent; 2 drivers	$\phi_y = .42, d_2 = .22$	$g \in [.20, .90];$ $\sigma = 1.5$
Train: direct_saturating	saturating / direct	$s = 6$	$g \in [.20, .90];$ $\sigma = 1.3$
Train: direct_threshold	threshold / direct	$\tau = 4, a_- = 1.30$	$g \in [.20, .90];$ $\sigma = 1.7$
Train: persistent_saturating	saturating / persistent	$\phi_y = .48, s = 7$	$g \in [.20, .90];$ $\sigma = 2.0$
Held mediated_feedback_linear	out: linear / mediated-feedback	$\phi_y = .24, \phi_m = .38, c = .68, f = .18$	$g \in [.20, .90];$ $\sigma = 1.8$
Held out: mediated_saturating	saturating / mediated	$\phi_y = .20, \phi_m = .44, c = .82, s = 6.5$	$g \in [.20, .90];$ $\sigma = 1.6$
Held out: persistent_threshold	threshold / persistent	$\phi_y = .58, \tau = 3.5, a_- = 1.20$	$g \in [.20, .90];$ $\sigma = 1.9$
Held feedback_saturating_extrap	out: saturating / feedback	$\phi_y = .44, \phi_m = .40, f = .32, s = 8$	$g \in [1.02, 1.28];$ $\sigma = 4.2$

All outcomes are clipped to $[0, 100]$. In table order, the world-specific baselines are 48, 55, 44, 51, 58, 47, 52, 46 for training and 54, 43, 57, 49 for held out. Thus the extrapolation block places both gain and observation noise strictly beyond the training hull ($g \leq .90, \sigma \leq 2.0$), while the other three held-out blocks isolate unseen compositions at in-hull parameters.

Table 34: **Structural-ODD training and evaluation protocol.** The two confirmatory reward arms receive byte-identical prompts and differ only in their training target.

Quantity	Value
Model roster	Qwen/Qwen3-4B-Instruct-2507, Qwen/Qwen3-8B, and meta-llama/Llama-3.1-8B-Instruct.
Chat formatting	Qwen3-4B uses its explicit ChatML wrapper; Qwen3-8B uses its tokenizer chat template with thinking disabled; Llama uses its tokenizer chat template. All add an assistant-generation prefix.
Roster-contrast scope	The Qwen comparison is a within-family checkpoint/scale contrast and the Qwen3-8B–Llama-3.1-8B comparison is approximately parameter matched. Neither identifies a pure scale or architecture effect because post-training release and tokenizer formatting also differ.
Confirmatory grid	2 disclosures $\times$ 3 models $\times$ 2 arms (causal.family, population.prior) $\times$ 3 seeds (42, 43, 44) = 36 training runs.
Diagnostic and bases	Qwen3-4B structureless control under both disclosures and all three seeds (6 runs), plus all 3 untrained bases under both disclosures (6 evaluation runs); total synthetic design 48 runs.
Training data	8 worlds $\times$ 600 episodes = 4, 800 rows per disclosure/arm; three calibration trajectories of length 10 and target evidence depth $k \in \{3, 6, 9\}$, balanced round-robin.
Held-out data	4 structural-ODD worlds $\times$ 60 episode seeds $\times$ 3 evidence depths = 720 paired rows. Training seed bands begin at 31,000,000 and evaluation bands at 73,000,000; they are disjoint.
Interventions	Shocks $\{-12, -4, 0, 4, 12\}$ at horizons 1 and 3, yielding ten joint forecasts and eight nonzero-minus-zero response contrasts.
Output contract	The complete response must be exactly one JSON object $\{"forecasts": [x_A, \dots, x_J]\}$ with one key and ten finite numbers. The prompt specifies the opening and closing literals separately and contains no non-JSON angle-bracket placeholder object. Extra prose/keys, duplicate keys, booleans, non-finite values, and wrong lengths are invalid under the same parser in online reward and offline scoring.
Structured decoding	A syntax-only GBNF constraint is applied identically during training rollouts, online greedy evaluation, endpoint greedy/stochastic evaluation, and untrained-base evaluation. It enforces the single key and ten-number arity but places no range or target-dependent constraint on any number. There is no parser repair or model-specific exception.

Structural-OOD training and evaluation protocol (continued).

Quantity	Value
Causal-family reward	$.4 r_{\text{level}} + .6 r_{\text{response}}$, where each level term is $\max(0, 1 - \hat{y} - y /10)$ and each response term is $\max(0, 1 - \widehat{\Delta y} - \Delta y /4)$, averaged over forecasts or contrasts.
Comparator targets	<code>population_prior</code> uses the frozen catalog-wide response vector; <code>structureless</code> independently resamples displayed drivers and uses a flat intervention response. Evaluation targets are always the true held-out mechanism and are byte-identical across training arms.
Full training	4,800 presentations, 300 rollout-and-update rounds, 2,400 AdamW steps; online greedy evaluation every 25 rounds; final adapter retained.
Online evaluation	All 720 rows; temperature 0, top- $p = 1$, one draw, 192-token limit. Parse failures receive the full observable-range error penalty rather than being dropped.
Endpoint decoding	Greedy retained for continuity. Additionally, five draws per episode at temperature .7, top- $p = .95$, 192-token limit; draws are averaged within episode. Untrained bases use decode seeds 20260814 (greedy) and 20260815 (stochastic); trained stochastic seed is training seed +20,260,814.
Variance estimation	5,000 hierarchical bootstrap repetitions, RNG seed 20260814: resample the three training seeds at the top level and paired episodes within seed; repeated decodes are never counted as independent training replicates.
Registered contrasts	The primary contrast is population-prior minus causal-family response MAE, pooled equally over all 720 held-out episodes. The same contrast within each of the four held-out blocks is a prespecified stratified analysis. Secondary contrasts, specified before cross-model outcomes, are untrained base minus causal-family error, structureless minus causal-family error, the difference in causal-vs-prior benefit for Qwen3-8B versus Qwen3-4B and for Qwen3-8B versus Llama-3.1-8B, and the disclosed-minus-undisclosed difference in causal-vs-prior benefit. Positive values favor causal training or the first-named condition. All use the same seed/paired-episode bootstrap hierarchy.
Reporting rule	All six model-by-disclosure primary estimates and every registered secondary contrast are reported; none is selected by its result. Intervals are unadjusted 95% hierarchical intervals. Broad roster claims require directional consistency across models and both decoding analyses.
Preflight validation	For both disclosures and all three models, a 10-round seed-42 run must save an adapter; greedy and stochastic strict-parse rates must each be at least .95; reward SD must exceed 10^{-4} ; and the maximum per-round fraction of length-limited generations must not exceed .05. Each untrained base must cover all four held-out blocks at $k \in \{3, 6, 9\}$, yielding 720 greedy and 3,600 stochastic scored draws per model/disclosure cell with both parse rates at least .95. Parse failures remain penalized forecasts in the final analysis.
Compute	Two NVIDIA A6000 48-GB GPUs per training run: one vLLM actor and one DeepSpeed learner, non-allocated; 8 CPU cores and 100 GB host memory. Base and endpoint-only evaluations use one A6000, 8 CPU cores, and 24 GB host memory.

Table 35: **Complete 48-endpoint structural-OOD comparison.** The grid contains 42 trained endpoints and six untrained bases. Base, Prior, and Causal are response MAE pooled equally over all 720 held-out episodes; $\Delta_{\text{causal}} = \text{Prior} - \text{Causal}$, so positive values favor causal-family training. Stochastic rows average five draws within episode before the seed-first bootstrap.

Decode	Disclosure	Model	Base	Prior	Causal	Δ_{causal}	[95% CI]	Mech. acc.
Greedy	Disclosed	Qwen3-4B	2.57	3.51	3.11	0.40	[-0.22,0.98]	0.270
Greedy	Disclosed	Qwen3-8B	2.63	1.48	1.65	-0.17	[-0.39,0.01]	0.270
Greedy	Disclosed	Llama-3.1-8B	3.26	2.37	2.15	0.22	[-0.17,0.62]	0.283
Greedy	Undisclosed	Qwen3-4B	3.25	3.94	3.65	0.29	[-0.10,0.76]	0.281
Greedy	Undisclosed	Qwen3-8B	2.99	2.50	2.19	0.31	[-0.20,0.83]	0.280
Greedy	Undisclosed	Llama-3.1-8B	4.14	3.18	2.48	0.70	[0.12,1.26]	0.288
Stochastic	Disclosed	Qwen3-4B	2.53	1.81	1.48	0.32	[0.26,0.42]	0.265
Stochastic	Disclosed	Qwen3-8B	2.56	1.51	1.63	-0.12	[-0.21,-0.03]	0.251
Stochastic	Disclosed	Llama-3.1-8B	3.83	2.16	2.12	0.04	[-0.61,0.88]	0.318
Stochastic	Undisclosed	Qwen3-4B	3.26	1.82	1.48	0.34	[0.31,0.37]	0.292
Stochastic	Undisclosed	Qwen3-8B	3.20	1.57	1.58	-0.01	[-0.06,0.03]	0.260
Stochastic	Undisclosed	Llama-3.1-8B	3.90	2.48	1.39	1.09	[0.37,2.40]	0.296

The primary contrast is model-dependent. Under repeated sampling, causal-family training beats the population-prior arm for Qwen3-4B with the mechanism shown (0.32, 95% interval [0.26, 0.42]) and hidden (0.34 [0.31, 0.37]). For Qwen3-8B, the disclosed contrast reverses (−0.12 [−0.21, −0.03]) and the hidden contrast is unresolved (−0.01 [−0.06, 0.03]). Llama-3.1-8B is unresolved with disclosure (0.04 [−0.61, 0.88]) but favors causal training without it (1.09 [0.37, 2.40]).

Greedy decoding favors causal training only for the undisclosed Llama cell (0.70 [0.12, 1.26]). The complete roster therefore does not support a model-general causal-over-prior advantage.

Table 36: **Registered secondary contrasts for the complete structural-OOD grid.** Positive Base—causal and Structureless—causal values favor causal-family training. Model rows compare the causal-over-prior benefit; disclosure rows compare that benefit when the mechanism text is shown versus hidden.

Contrast	Scope	Greedy [95% CI]	Stochastic [95% CI]
Base — causal	Disclosed / Qwen3-4B	-0.54 [-1.10,-0.02]	1.05 [0.96,1.14]
Base — causal	Disclosed / Qwen3-8B	0.98 [0.76,1.16]	0.93 [0.79,1.09]
Base — causal	Disclosed / Llama-3.1-8B	1.11 [0.66,1.52]	1.71 [1.25,2.25]
Base — causal	Undisclosed / Qwen3-4B	-0.41 [-1.09,0.20]	1.79 [1.71,1.86]
Base — causal	Undisclosed / Qwen3-8B	0.79 [0.34,1.17]	1.61 [1.40,1.87]
Base — causal	Undisclosed / Llama-3.1-8B	1.66 [1.05,2.28]	2.51 [2.44,2.57]
Structureless — causal	Disclosed / Qwen3-4B	0.82 [0.24,1.36]	0.64 [0.55,0.78]
Structureless — causal	Undisclosed / Qwen3-4B	1.07 [-0.05,2.08]	0.68 [0.57,0.86]
Qwen3-8B — Qwen3-4B causal benefit	Disclosed	-0.57 [-1.22,0.11]	-0.45 [-0.59,-0.30]
Qwen3-8B — Llama-3.1-8B causal benefit	Disclosed	-0.39 [-0.78,-0.01]	-0.16 [-1.07,0.49]
Qwen3-8B — Qwen3-4B causal benefit	Undisclosed	0.02 [-0.62,0.70]	-0.35 [-0.40,-0.31]
Qwen3-8B — Llama-3.1-8B causal benefit	Undisclosed	-0.39 [-1.07,0.31]	-1.11 [-2.43,-0.34]
Disclosed — undisclosed causal benefit	Qwen3-4B	0.11 [-0.50,0.72]	-0.02 [-0.09,0.10]
Disclosed — undisclosed causal benefit	Qwen3-8B	-0.48 [-1.08,0.11]	-0.11 [-0.22,0.04]
Disclosed — undisclosed causal benefit	Llama-3.1-8B	-0.48 [-1.24,0.31]	-1.06 [-2.54,0.51]

Training improves on the untrained endpoint under repeated sampling, but reward-arm effects remain heterogeneous. Causal-family training lowers stochastic response MAE relative to the untrained base in all six model-by-disclosure cells; the registered contrasts range from 0.93 to 2.51, and every interval excludes zero. In the Qwen3-4B diagnostic, it also beats structureless training with disclosure (0.64 [0.55, 0.78]) and without it (0.68 [0.57, 0.86]). The Qwen scale contrasts are negative under repeated sampling in both disclosure conditions, so increasing Qwen parameter count does not increase the causal-over-prior benefit in this grid.

Disclosing the mechanism does not reliably change the training contrast. For every model and decoding regime, the interval for the disclosed-minus-hidden causal benefit includes zero. Overall exact mechanism accuracy for causal models ranges from .251 to .318, near the four-way chance rate. The block tables show that this average is concentrated: extrapolation accuracy is near one while accuracy on topology and nonlinear compositions is near zero. Lower forecast error therefore does not imply uniform identification of the held-out mechanism.

Table 37: **Greedy structural-OOD results by held-out block.** $\Delta_{\text{causal}} = \text{Prior} - \text{Causal}$; response MAE assigns the fixed 100-point penalty to parse failures.

Disclosure	Model	Held-out block	Base	Prior	Causal	Δ_{causal}	[95% CI]	Mech. acc.
Disclosed	Qwen3-4B	topology	2.05	2.21	1.98	0.23	[-0.64,0.80]	0.017
Disclosed	Qwen3-4B	nonlinear	1.34	3.37	2.46	0.91	[0.20,1.99]	0.004
Disclosed	Qwen3-4B	mixed	2.29	2.03	1.88	0.15	[-0.59,0.89]	0.198
Disclosed	Qwen3-4B	extrapolation	4.59	6.45	6.12	0.33	[-1.71,2.46]	0.863
Disclosed	Qwen3-8B	topology	2.23	1.60	1.41	0.19	[-0.40,0.45]	0.022
Disclosed	Qwen3-8B	nonlinear	1.49	1.05	0.80	0.25	[0.19,0.30]	0.030
Disclosed	Qwen3-8B	mixed	2.46	1.51	1.40	0.11	[0.06,0.16]	0.059
Disclosed	Qwen3-8B	extrapolation	4.34	1.76	2.99	-1.23	[-1.88,-0.84]	0.969
Disclosed	Llama-3.1-8B	topology	2.89	1.79	1.49	0.30	[0.28,0.33]	0.000
Disclosed	Llama-3.1-8B	nonlinear	1.66	2.12	1.57	0.55	[-0.18,1.29]	0.000
Disclosed	Llama-3.1-8B	mixed	3.93	2.87	3.46	-0.59	[-1.83,0.55]	0.187
Disclosed	Llama-3.1-8B	extrapolation	4.55	2.69	2.08	0.62	[0.04,1.51]	0.944
Undisclosed	Qwen3-4B	topology	2.90	2.96	3.13	-0.17	[-1.19,0.72]	0.046
Undisclosed	Qwen3-4B	nonlinear	1.72	4.77	3.83	0.94	[-0.12,2.23]	0.011
Undisclosed	Qwen3-4B	mixed	2.84	2.95	3.12	-0.17	[-0.71,0.04]	0.135
Undisclosed	Qwen3-4B	extrapolation	5.52	5.09	4.53	0.55	[0.16,1.16]	0.930
Undisclosed	Qwen3-8B	topology	2.67	3.11	2.01	1.10	[-0.07,2.38]	0.033
Undisclosed	Qwen3-8B	nonlinear	1.66	2.01	1.23	0.78	[0.22,1.50]	0.017
Undisclosed	Qwen3-8B	mixed	2.53	2.62	2.31	0.31	[-0.92,1.54]	0.085
Undisclosed	Qwen3-8B	extrapolation	5.08	2.27	3.21	-0.94	[-1.62,-0.47]	0.983
Undisclosed	Llama-3.1-8B	topology	4.97	3.83	2.18	1.64	[0.44,2.88]	0.006
Undisclosed	Llama-3.1-8B	nonlinear	2.62	3.02	2.48	0.54	[-0.81,1.90]	0.004
Undisclosed	Llama-3.1-8B	mixed	3.95	3.25	3.16	0.09	[-0.63,0.81]	0.169
Undisclosed	Llama-3.1-8B	extrapolation	5.02	2.60	2.08	0.53	[-0.21,1.22]	0.976

Table 38: **Five-draw stochastic structural-OOD results by held-out block.** Draws are averaged within episode; all other conventions match Table 37.

Disclosure	Model	Held-out block	Base	Prior	Causal	Δ_{causal} [95% CI]	Mech. acc.
Disclosed	Qwen3-4B	topology	1.99	1.86	1.37	0.48 [0.43,0.53]	0.007
Disclosed	Qwen3-4B	nonlinear	1.31	1.43	0.94	0.49 [0.46,0.54]	0.003
Disclosed	Qwen3-4B	mixed	2.22	1.14	1.15	-0.01 [-0.06,0.05]	0.190
Disclosed	Qwen3-4B	extrapolation	4.60	2.79	2.47	0.32 [0.02,0.77]	0.860
Disclosed	Qwen3-8B	topology	2.06	1.50	1.11	0.39 [0.34,0.44]	0.000
Disclosed	Qwen3-8B	nonlinear	1.39	1.04	0.72	0.32 [0.27,0.37]	0.020
Disclosed	Qwen3-8B	mixed	2.34	1.28	1.30	-0.02 [-0.09,0.04]	0.030
Disclosed	Qwen3-8B	extrapolation	4.45	2.21	3.39	-1.18 [-1.40,-0.95]	0.954
Disclosed	Llama-3.1-8B	topology	3.21	1.69	1.39	0.29 [0.23,0.37]	0.000
Disclosed	Llama-3.1-8B	nonlinear	3.49	3.11	3.63	-0.52 [-3.08,2.99]	0.004
Disclosed	Llama-3.1-8B	mixed	3.43	1.16	1.05	0.11 [0.05,0.17]	0.297
Disclosed	Llama-3.1-8B	extrapolation	5.21	2.67	2.41	0.26 [0.12,0.42]	0.971
Undisclosed	Qwen3-4B	topology	2.81	1.87	1.34	0.53 [0.49,0.58]	0.030
Undisclosed	Qwen3-4B	nonlinear	1.70	1.44	0.92	0.52 [0.48,0.55]	0.001
Undisclosed	Qwen3-4B	mixed	2.80	1.14	1.11	0.03 [-0.01,0.07]	0.177
Undisclosed	Qwen3-4B	extrapolation	5.73	2.82	2.53	0.28 [0.17,0.38]	0.958
Undisclosed	Qwen3-8B	topology	2.37	1.37	1.11	0.26 [0.23,0.29]	0.003
Undisclosed	Qwen3-8B	nonlinear	1.72	1.03	0.75	0.28 [0.26,0.31]	0.004
Undisclosed	Qwen3-8B	mixed	2.48	1.21	1.25	-0.04 [-0.08,0.00]	0.063
Undisclosed	Qwen3-8B	extrapolation	6.23	2.68	3.24	-0.55 [-0.79,-0.37]	0.971
Undisclosed	Llama-3.1-8B	topology	3.09	1.66	1.38	0.28 [0.11,0.43]	0.014
Undisclosed	Llama-3.1-8B	nonlinear	3.39	4.07	0.94	3.13 [0.43,7.72]	0.001
Undisclosed	Llama-3.1-8B	mixed	3.47	1.16	1.06	0.10 [0.06,0.14]	0.197
Undisclosed	Llama-3.1-8B	extrapolation	5.64	3.05	2.18	0.87 [0.56,1.16]	0.970

Structured-output validity. All synthetic arms use one syntax-only GBNF decoder because malformed JSON would otherwise mix forecasting error with a model-family formatting artifact. The grammar constrains only object shape and array length, permits EOS only after the root is complete, and places no restriction on numeric values. Parse failures remain maximally penalized forecasts in the endpoint analysis.

Table 39: **Runtime environment for the reported training runs.** Package versions come from the job execution environment.

Component	Version / configuration
Python / CUDA build	Python 3.10.20; PyTorch 2.6.0 built for CUDA 12.4.
Training stack	OAT 0.2.4 at Git revision 869706620a35ec5304fa80a0c74f0c566cc2bc57; DeepSpeed 0.16.8; PEFT 0.15.2.
Generation stack	Transformers 4.51.3; vLLM 0.8.4; FlashAttention 2.7.4.post1.
Data stack	Datasets 2.16.1; Accelerate 1.14.0.
Allocator / logging	PYTORCH_CUDA_ALLOC_CONF=expandable_segments:True; Weights & Biases disabled; stdout/stderr and per-round metrics retained with each run.

C.4 REPRODUCIBILITY

The artifact roots `exp3_training_transfer/mechanism_family/`, `exp3_training_transfer/coin_city_structural/`, and `polymarket/scripts/` contain the generators, prompt builders, trainers, evaluators, and analysis code for the three studies. Execution ledgers record effective hyperparameters, Git revisions, adapter identities, and SHA-256 hashes of data and executable protocol files; result renderers verify these fields before aggregation.

For structural OOD, `reports/c3_mechanism_full_aggregate.json` records the 48-job ledger, hashes for all 96 greedy and stochastic score files, 207,360 scored row-draws, the bootstrap seed, analysis-code hashes, and every reported contrast. For Coin City, `make_qwen3_4b_`

2862 `endpoint_outputs.py` validates the ten five-draw files (72,000 scored draws), performs the
2863 paired seed-first bootstrap, and writes Figure 7 and Table 25; the companion greedy analysis ver-
2864 ifies the round-300 task universe and writes Table 27. The Polymarket manifests preserve sample
2865 membership, prompt and adapter hashes, parse coverage, and bootstrap settings.
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